Graph neural networks recover interpretable circuit models from neural activity

Anonymous Author(s)
Affiliation
Address
email

Green: tables and figures regenerated with the general-form GNN (group lasso 25) of the 2026 experiment campaign. Red: a sentence whose result changed with the new analysis, or a section chosen for rewriting.

Abstract

Synapse-level connectomes describe the structure of neural circuits but not their dynamics; neural activity recordings capture dynamics but not structure. Recovering circuit parameters from activity alone is possible when both connectivity and activity are sufficiently rich and identifiable. This may fail on the sparse, repeat-column architectures of biological nervous systems, where parameter symmetries and limited state-space coverage can make additional constraints from connectivity or perturbations necessary to disambiguate the inverse problem. We explore conditions under which mechanistic inference from connectivity and activity is feasible, using a connectome-constrained simulator. We train graph neural networks (GNNs) on a simulation of the visual-system of the fruit fly *Drosophila melanogaster* with 13,741 neurons, 65 cell types, and up to 434,112 synaptic connections, driven by naturalistic video. Given the binary connectome graph and activity, the GNNs jointly recover the unknown per-neuron update functions and per-synapse interaction functions, together with effective connectivity weights, membrane time constants, resting potentials, and unsupervised cell-type clusters—on par with an oracle that knows the true ODE form. A SIREN extension additionally recovers unknown visual stimuli. Recovery remains robust when the connectome prior is relaxed by adding +400% random edges, and across FlyWire-derived connectome variants with heterogeneous wiring. We find that per-neuron noise breaks degeneracies that otherwise render the system unidentifiable, suggesting that perturbations enhancing data diversity are essential when activity alone is insufficient. ML baselines match one-step activity prediction but their learned dynamics do not encode the underlying connectivity neural dynamics, showing that fitting activity is a poor proxy for recovering mechanism.

1 Introduction

Building mechanistic, parameter-resolved models of neural circuits has been a long-standing goal of neuroscience. Small circuits have been modeled this way for decades, Hodgkin and Huxley's [1952] squid axon to the stomatogastric ganlion [Marder and Taylor, 2011]. Scaling these methods up has proven much harder: top-down, hand-specified large-scale simulations have struggled to be predictive of circuit function [Markram et al., 2015, Frégnac and Laurent, 2014]. Synapse-resolution connectomes that are paired with neural activity recordings make data-constrained mechanistic models conceivable in principle [The MICrONS Consortium, 2025]. In practice, however, neither connectivity nor activity will ever be complete or noise free: connectomes carry reconstruction errors and uncertain synaptic weights, and activity recordings cover fractions of neurons with measurement

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noise. The practical question is therefore not whether complete data will eventually arrive, but which combination of incomplete connectivity and activity is sufficient to recover circuit mechanism.

Three threads of recent work bound the answer but leave it open. *Foundation models of neural activity* trained on large-scale recordings achieve impressive cross-stimulus and cross-animal generalization [Wang et al., 2025], but are explicitly non-mechanistic. Their parameters do not map to biophysical quantitites, and counterfactual interventions on connectivity or cell types are not interpretable. *Connectome-constrained models* make real progress toward mechanism but pre-specify the dynamics: a fixed biophysical neuron model on the *C. elegans* connectome [Mi et al., 2021] or on the fly visual system [Lappalainen et al., 2024], or a linear effective-connectivity model on the fly brain Pospisil et al. [2024]. *In silico identifiability studies* probe the inverse problem directly [Yu et al., 2025, Lueckmann et al., 2026], but at small scale or without validating recovered connectivity. None of these directly characterize when mechanistic recovery is feasible at circuit scale, and none offers a flexible inference method that does not pre-specify the dynamical equation form.

Because no current experimental dataset provides complete synapse-resolution connectivity together with dense activity recordings from the same animal, we use Lappalainen et al.'s [2024] connectome-constrained Flyvis simulator as a proxy for paired data. The simulator is a biologically plausible continuous-time dynamical system on the real fly connectome, driven by naturalistic video, with known per-neuron and per-synapse parameters. We explore this system at different scales between 13K and 50K neurons, resulting in 0.4M to 5M edges, and heterogeneous wiring that we extracted from the FlyWire connectome [Dorkenwald et al., 2024, Schlegel et al., 2024] to test the convolutional assumption of the original model and connectivity-uncertainty. On this in silico ground truth, we evaluate a range of inference methods—ML sequence models, and oracle baseline that knows the true ordinary differential equation (ODE), and our primary contribution: an interpretable continuous-time GNN that does not assume the form of the dynamical equation. Instead, it learns the per-neuron update function and the per-synapse interaction function as shared MLPs conditioned on a learned per-neuron latent embedding, jointly with effective connectivity weights. Because the same functional forms must produce the diverse dynamics of 65 cell types, cell-type identity emerges in the latent embedding as a consequence of mechanistic recovery.

Across realistic data degradations, we identify several non-obvious regularities about when recovery is feasible. First, the GNN recovers the unknown update and synaptic-interaction functions from data alone, on par with a known-ODE oracle. Cell-type identity emerges unsupervised in the latent embedding space, with clustering accuracy matching that obtained from ground-truth parameters. Second, the connectome serves well as *binary* structural prior: the GNN tolerates a 400% inflation of the edge set with random null edges and still recovers true synaptic weights at $R^2 = 0.96$, while removing 50% of true edges breaks recovery. The practically useful connectome is therefore a superset of true edges, and knowing which pairs do *not* connect matters more than precise weights. Third, activity richness is the binding constraint on identification. Naturalistic stimulation alone leaves the inverse problem degenerate, and per-neuron noise is what breaks these degeneracies and allows the GNN to match the known-ODE oracle. We interpret model noise as one instance of the broader requirement that identification needs activity diversity beyond passive sensory drive, whether through intrinsic stochasticity or (by analogy and to be tested) through extrinsic perturbations such as targeted optogenetic stimulation. Fourth, prediction is not mechanism. ML sequence baselines reach one-step Pearson $r \geq 0.95$ on activity but cannot be used to probe mechanistic interventions such as ablating 50% of synaptic connections that both the GNN and the known-ODE oracle naturally permit. Finally, measurement noise on voltage traces remains the principal open bottleneck: a $\gamma = 0.1$ measurement noise leaves rollouts intact but degrades synaptic-weight recovery from $R^2 = 0.99$ to 0.63, and we flag this as the central challenge for connectome-inverse methods on real data.

Our contributions are: (1) an interpretable continuous-time GNN that recovers the per-neuron update function and per-synapse interaction function from connectivity and activity, without assuming the form of the dynamical equation. Cell-type identity emerges in the learned latent embedding as a structural consequence; the framework extends with a SIREN module to additionally recover unobserved sensory inputs; (2) a characterization of identifiability across realistic data degradations on a *Drosophila* visual-system simulator, including FlyWire heterogeneous wiring and false-positive connectivity; (3) an agentic hyperparameter-optimization protocol that yielded a single training configuration robust across conditions; and (4) the finding that perturbation-driven activity richness, rather than architectural sophistication, is the binding constraint on mechanistic recovery from connectivity and activity.

2 Related work

Lueckmann et al. [2026] establish an in silico testbed for connectome-constrained system identification on a larval zebrafish visuomotor circuit of 36 units, showing that LLM-guided code search can recover effective connectivity when given structural priors. The setup is closest to ours in spirit, and complementary in scope: their circuit is small enough to admit symbolic search, ours is two orders of magnitude larger and forces a parameterised inference method. Lappalainen et al. [2024] demonstrated that a connectome-constrained recurrent network with a hand-designed ODE on the *Drosophila* visual system can recapitulate plausible single-neuron physiology after training on optical flow. We use their simulator as in silico ground truth precisely because of this evidence of biological plausibility, and ask the inverse question: can the underlying dynamical system be *recovered* from activity and connectivity, without being told its functional form?

Connectome-constrained inference with a fixed biophysical neuron model has been demonstrated on *C. elegans* calcium data [Mi et al., 2021] and via linear effective-connectivity from perturbations on the *Drosophila* connectome [Pospisil et al., 2024]; both pre-specify the dynamics. Foundation models of activity [Wang et al., 2025] achieve strong cross-stimulus and cross-animal prediction on $\sim 10^5$ real neurons but do not target mechanism. Symbolic-regression and equation-discovery approaches [Yu et al., 2025, Prasse and Van Mieghem, 2022] recover dynamical equations from simulated activity but do not validate connectivity weights or learn neuronal heterogeneity. None of these directly characterises the conditions under which mechanistic recovery from connectivity and activity is feasible.

Our GNN inherits its decomposition principle from Allier et al.'s [2024, 2026] work on random recurrent neural assemblies. However, recovering the parameters of a biological circuit on a real, sparsely connected, repeated-column connectome is a different problem. Neural assemblies with random weights do not exhibit the structural degeneracies that columnar visual systems do: when same-type neurons across columns produce time-shifted copies of one another's signals, large families of distinct synaptic weight matrices generate indistinguishable activity. We characterise these degeneracies analytically (Appx. C) and find, for instance, that a circuit retaining only 29% of the original edges produces activity indistinguishable from the full circuit under naturalistic stimulation. This motivates the methodological additions of this work: a Known-ODE oracle that quantifies the residual gap between data-driven inference and full equation knowledge, an edge-ablation mechanistic test, the identification of per-neuron noise as a perturbation that breaks columnar degeneracies, and an agentic hyperparameter-optimisation procedure that produces a single training configuration robust across conditions of connectivity completeness and activity quality.

3 Methods

Flyvis simulations. We simulated neural activity in the fly visual system using Flyvis pretrained models. Each neuron is modeled as a non-spiking voltage compartment governed by the ODE

$$\tau_i \frac{dv_i(t)}{dt} = -v_i(t) + V_i^{\text{rest}} + \sum_{j \in \mathcal{N}_i} W_{ij}\, \text{ReLU}\big(v_j(t)\big) + I_i(t) + \sigma\, \xi_i(t), \tag{1}$$

where v_i is the neuron membrane voltage, τ_i and V_i^{rest} are cell-type parameters, W_{ij} is the connectome synaptic weight, and $I_i(t)$ is the visual input. Visual inputs are derived from the DAVIS-2017 benchmark dataset [Perazzi et al., 2016] for annotated natural videos. A null stimulus ($I_i(t) = 0$) is prepended to every sequence so that neurons relax toward V_i^{rest} during the first half of each clip, which is essential for parameter identifiability (Appx. D). $\xi_i(t) \sim \mathcal{N}(0, 1)$ is independent Gaussian noise scaled by σ (model noise), perturbing the dynamical trajectory of $v_i(t)$. We additionally apply measurement Gaussian noise scaled by γ directly to $v_i(t)$ (Supp. Figs. 3 and 4). We first restricted the original 721 retinotopic columns to the central subset of 217 columns (Flyvis-217), covering the central visual field while substantially reducing memory and runtime requirements during tests and optimization. This simulated recurrent neural network contains 13,741 neurons from 65 cell types and 434,112 synaptic connections. Ground-truth distributions of the per-neuron parameters τ_i, V_i^{rest} and per-edge weights W_{ij} across cell types are reported in Supp. Fig. 5. Next, we expanded the simulations to a hybrid connectome that retains the trained Flyvis dynamical model but inherits per-column wiring heterogeneity and connectivity uncertainty from FlyWire, resulting in networks with 50,412 neurons and up to 9,642,335 edges (Appx. E and Supp. Tab. 3). We evaluate these

networks in two configurations, an 8 column extent central region corresponding to Flyvis-217 (e8 hybrid) and the full eye (FlyWire eye).

Graph neural network model We approximate the simulated voltage dynamics by a message-passing GNN [Gilmer et al., 2017]

$$\frac{d\hat{v}_i(t)}{dt} = f_\theta\Big(v_i(t),\, \mathbf{a}_i,\, \sum_{j\in\mathcal{N}_i} \hat{W}_{ij}\, g_\phi\big(v_j(t), \mathbf{a}_j\big)^2,\, I_i(t)\Big). \tag{2}$$

Nodes of the GNN correspond to neurons and edges correspond to functional synaptic connections. The GNN learns a latent embedding $\mathbf{a}_i \in \mathbb{R}^2$ for each neuron. The learnable vectors $\mathbf{a}_i \in \mathbb{R}^2$ represent compact latent identity, they can map to cell-type-specific properties (like time constants and nonlinearities). Neuron update $f_\theta = \text{MLP}_0$ and edge message $g_\phi = \text{MLP}_1$ are three-layer perceptrons (width 80, ReLU, linear output). g_ϕ maps presynaptic inputs v_j to nonnegative messages (via squaring) depending on latent embedding $\mathbf{a}_j$, weighted by $\hat{W}_{ij}$. f_θ processes the postsynaptic voltage v_i, aggregated input, and external input $I_i(t)$ to predict $d\hat{v}_i(t)/dt$, depending on $\mathbf{a}_i$. During training, inputs $I_i(t)$, adjacency $\mathcal{N}_i$, and activity $v_i(t)$ are given (Appx. F.1). The MLPs, $\hat{W}_{ij}$, and $\mathbf{a}_i$ are optimized by minimizing the difference between simulator $y_i(t) = dv_i(t)/dt$ and GNN predictions $\hat{y}_i(t) = d\hat{v}_i(t)/dt$:

$$\mathcal{L}_{\text{pred}} = \sum_{i,t} \|\hat{y}_i(t) - y_i(t)\|_2 \tag{3}$$

Many synaptic weight configurations give rise to identical neural activity, making the recovery of $\hat{\mathbf{W}}$ from voltage traces fundamentally non-unique (Appx. C). To constrain the solution space, we added regularization terms to $\mathcal{L}_{\text{pred}}$

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{pred}} &+ \lambda_0\|\theta\|_1 + \lambda_1\|\phi\|_1 + \lambda_2\|\theta\|_2 + \gamma_0\|\hat{\mathbf{W}}\|_1 + \gamma_1\|\hat{\mathbf{W}}\|_2 \\ &+ \mu_1\Big\|\text{ReLU}\Big(-\tfrac{\partial\, g_\phi(v_j,\mathbf{a}_j,v_i,\mathbf{a}_i)}{\partial v_j}\Big)\Big\|_2 + \mu_2\|g_\phi(v_\star,\mathbf{a}_j,v_\star,\mathbf{a}_i) - v_\star\|_2 + \nu\textstyle\sum_k \|\phi_k^{(1)}\|_2.\end{aligned} \tag{4}$$

The ℓ_1 penalties λ_0, λ_1 on the MLP weights θ and ϕ promote sparsity, biasing f_θ and g_ϕ toward simpler functions; the ℓ_2 penalty λ_2 on θ prevents weight explosion and stabilizes training. The ℓ_1 term γ_0 on $\hat{\mathbf{W}}$ promotes sparse effective connectivity, while the ℓ_2 term γ_1 penalizes large synaptic weights. [The published monotonicity term μ_0 on f_θ and the ℓ_2 term λ_3 on ϕ are zero in the general-form GNN runs and are left out of Eq. (4).] The monotonicity term μ_1 penalizes negative slopes of g_ϕ with respect to the presynaptic voltage v_j, encoding the prior that synaptic drive increases with presynaptic voltage. The normalization term μ_2 anchors g_ϕ at a reference voltage $v_\star$, removing the scale degeneracy between g_ϕ and $\hat{\mathbf{W}}$ in Eq. (2). [For the general-form GNN, $g_\phi(v_j, \mathbf{a}_j, v_i, \mathbf{a}_i)$ is evaluated for μ_1 and μ_2 at randomly sampled inputs: each neuron i is paired with a random partner j, redrawn at every iteration, irrespective of the connectome, so $(\mathbf{a}_i, \mathbf{a}_j)$ is not an edge of the graph; $v_i = v_j$ is neuron i's observed voltage.] The group lasso ν sums, over the four inputs of g_ϕ, the ℓ_2 norm of the columns $\phi_k^{(1)}$ of its first layer that read input k, so that an input g_ϕ does not need is removed as a whole. We tuned all the hyperparameters $\lambda_0, \lambda_1, \lambda_2, \gamma_0, \gamma_1, \mu_1, \mu_2, \nu$ and the MLPs' structure with an agentic optimization procedure (Appx. G) [HPO not run on the general-form GNN]. After training, the circuit parameters are extracted from the GNN as described in Appx. F.2.

Benchmark We compare GNN results with four baseline models detailed in Appx. H: Known-ODE, which assumes knowledge of the full functional form of the ODE and learns the circuit parameters τ, V^{rest}, W; Recurrent MLP, which predicts dv/dt as a function of the current $v(t)$ and $I(t)$ using a 2-layer MLP (640 hidden units) with ResNet-style skip connections; EED model, which predicts neural activity using a 256D nonlinear dynamical system that is read out by the observed activity; and Stimulus-driven MLP, which predicts $v(t)$ based only on the past stimulus $I(t-(w-1)), \cdots, I(t-1), I(t)$. These baselines span the spectrum from full mechanistic knowledge (Known-ODE) to purely data-driven prediction (Stimulus-driven MLP); Recurrent MLP is a general first-order dynamical system of voltages, and the EED model assumes a lower-dimensional latent dynamical system that is read out by the observed voltages.

4 Results

We report four quantities throughout the results section: the one-step and held-out auto-regressive rollout (7,207 to 7,999 frames by fold) [5-fold cross-validation: each fold simulates its training data

Table 1: **GNN and baselines on Flyvis-217** across three model-noise levels ($\sigma \in \{0, 0.05, 0.5\}$); 5-fold CV (mean $\pm$ SD). Prediction metrics on noise-free held-out stimuli. Parameter recovery (possible only for Known-ODE and GNN): $R^2_{\hat{W}}$ over the fitted edges of the 434,112 edges, outlier-corrected; $R^2_{\hat{\tau}}$ and $R^2_{\hat{V}^{\text{rest}}}$ over all 13,741 neurons, outlier-corrected (Appendix I); GMM clustering accuracy over 65 cell types. Green: > 0.9. (At $\sigma = 0.5$ the rollout from the initial voltages diverged ($r < 0.5$) in 2 of the 10 folds we trained, cv04 and cv09; the GNN row uses cv00–cv03 and cv05, cv04 replaced by the next fold that is not one.)

		model	prediction		parameter recovery			
model	condition	noise σ	one-step r	rollout r	$R^2_{\hat{W}}$ (out.%)	$R^2_{\hat{\tau}}$ (out.%)	$R^2_{\hat{V}^{\text{rest}}}$ (out.%)	cluster acc.
Known ODE	noise-free	0	1.00±0	1.00±0	0.97±6E−4 (0%)	1.00±3E−5 (2%)	0.90±2E−3 (2%)	0.92±2E−3
	low noise	0.05	1.00±0	1.00±0	0.99±2E−4 (0%)	1.00±8E−6 (0%)	0.95±1E−3 (0%)	0.93±5E−3
	high noise	0.5	1.00±0	1.00±0	1.00±2E−6 (0%)	1.00±7E−7 (0%)	1.00±9E−6 (0%)	0.93±5E−3
GNN (ours)	noise-free	0	1.00±0	1.00±0	0.92±0.01 (0%)	0.90±0.04 (4%)	0.87±0.02 (20%)	0.89±0.01
	low noise	0.05	1.00±0	1.00±4E−4	0.98±2E−3 (0%)	0.98±4E−3 (0%)	0.89±0.02 (3%)	0.90±0.01
	high noise	0.5	0.99±7E−4	0.99±1E−3	0.99±6E−4 (0%)	1.00±1E−3 (0%)	0.91±0.02 (1%)	0.88±0.02
Recurrent MLP	noise-free	0	0.95±0.04	0.96±0.05	—	—	—	—
	low-noise	0.05	0.92±0.07	0.93±0.11	—	—	—	—
	high-noise	0.5	0.69±0.18	0.78±0.24	—	—	—	—
EED Model	noise-free	0	0.92±0.08	0.97±0.03	—	—	—	—
	low-noise	0.05	0.90±0.10	0.97±0.04	—	—	—	—
	high-noise	0.5	0.60±0.20	0.77±0.25	—	—	—	—

(64,000 frames) from 48 of the 61 DAVIS videos and its held-out test data from the other 13, and uses its own simulation and training seeds] Pearson r on neural activity (per neuron, Fisher-z pooled), and the identity-line R^2 on $\hat{\mathbf{W}}$, $\hat{\boldsymbol{\tau}}$, and $\hat{\mathbf{V}}^{\text{rest}}$. The two parameter R^2's are computed on the inlier set defined by hard residual thresholds, with the outlier fraction shown in parentheses. Full definitions are given in Appx. I.

Identifiability of parameters. The recovery of circuit parameters from noise free simulation traces of the central Flyvis-217 model is ill-posed, even with 64,000 frames of diverse visual stimuli. In Appx. C, we rewrote the ODE into a linear form, and show that, even with full knowledge of the underlying equations, $\approx 10\%$ of the circuit parameters cannot be uniquely determined from neural activity alone (see Supp. Fig. 12). The dynamics of neurons that receive no presynaptic inputs, are continuously at rest, or whose outputs are always sub-threshold cannot be identified. For instance, we found a variant circuit with only 125,952 edges that generates almost the same activity as the complete 434,112 edges Flyvis model (rollout $r = 0.95$ over 1,000 frames).

Prediction but no circuit parameters. The neuronal activity of different cells encodes several frames of past stimuli so that the optic lobe can perform complex visual tasks. A Stimulus-driven MLP (8M parameters) is able to reproduce this neural activity from a fixed window of past stimulus (Supp. Tab. 8). Its one-step prediction on held-out frames steadily increases with the window size, reaching $r = 0.92 \pm 0.09$ with 8 context frames and $r = 0.97 \pm 0.03$ with 16. The Recurrent MLP (29M parameters) and the EED model (20M parameters) are accurate one-step predictors ($r > 0.9$) and produce faithful 8,000-frame autoregressive rollouts ($r > 0.95$) in the noise-free regime (Tab. 1). But, the performance degrades with the addition of model noise and we speculate that the models memorize noise patterns in the training data that do not generalize. Despite being predictors of neuron activity, these models *do not* learn the true underlying dynamics. Supp. Fig. 18 shows the discrepancy between the expected and learned dynamics visualized through analysis of their Jacobian. The learned dynamics are therefore not interpretable, there is no mapping from the model's parameters to Flyvis-217 model quantities (Appx. H).

Results with and without the ODE given. Predicting activity alone does not yield understanding of the underlying circuit. Mechanistic interpretation requires recovering the governing equation of the neural signal together with per-neuron and connection parameters. If we know the differentiable ODE (Eq. (1), 0.46M parameters), we can directly learn the connectivity weights $\hat{\mathbf{W}}$ and the per-neuron $\hat{\boldsymbol{\tau}}$, $\hat{\mathbf{V}}^{\text{rest}}$. The shared ReLU activation is known, but cell-type assignments are not. With ℓ_1/ℓ_2 regularisation on the parameters and an MSE one-step prediction loss (Appx. H), this trained Known-ODE models achieves near-perfect activity prediction (one-step r and rollout r are close to 1.00 over the held-out 7,207 to 7,999 frames. Notably, the Known-ODE trained on noisy activity reproduces the noise-free rollout: it cannot explicitly fit the stochastic fluctuations and instead converges on the underlying deterministic dynamics. The parameter recovery over $> 4.5 \cdot 10^5$ values is comparably

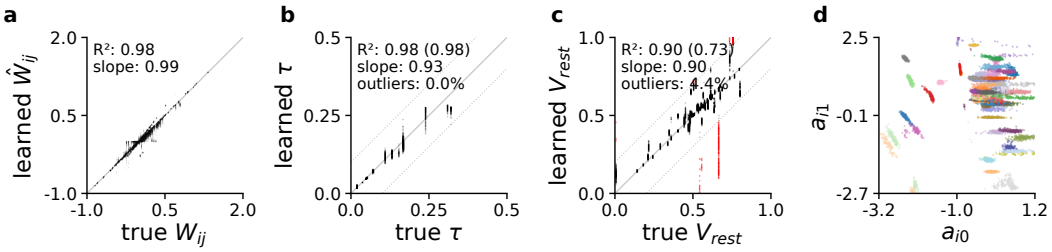


Figure 1: **Circuit parameter extraction** from the general-form GNN evaluated on Flyvis-217 at model noise $\sigma = 0.05$. **(a)** Connectivity weights $\hat{W}_{ij}$ over the fitted edges. **(b)** Time constants τ_i. **(c)** Resting potentials V_i^{rest}. **(d)** Learned latent embeddings $\mathbf{a}_i \in \mathbb{R}^2$ of all neurons. Each scatter is annotated with R^2 and slope of learned vs. true; colours indicate ground-truth cell types.

accurate even at high model noise ($R^2 > 0.95$, outliers $< 10\%$, Supp. Figs. 14 and 15). Although the inverse problem is strongly ill-posed, full knowledge of the equation form combined with ℓ_1/ℓ_2 regularisation makes it tractable. In contrast to Know-ODE, the GNN (0.48M parameters) does not have access to the update equation. It learns jointly the connectivity, a per-neuron embedding, and two signalling functions f_θ, g_ϕ that together express a wide family of update rules (Eq. (2)). The training scheme involves a large hyperparameter set (architecture, learning rates, regularisation coefficients) too large for grid search, so we tuned it with an agentic loop (Appx. F.1 and G). (Across ten trained folds the rollout from the initial voltages diverged ($r < 0.5$) in 2 folds at $\sigma = 0.5$, in 2 with $+400\%$ null edges and in 3 with the unknown stimulus, whereas the one-step prediction of those folds is normal ($r \geq 0.99$) and so is their parameter recovery, except in one unknown-stimulus fold; the tables report five folds, each outlier fold replaced by the next fold that is not one.) At low and high model noise ($\sigma \in \{0.05,\ 0.5\}$), the GNN results closely matches the Known-ODE upper bound. The deterministic $\sigma = 0$ regime is the harder case ($r = 0.90 \pm 0.21$, $R^2_{\hat{\mathbf{W}}} = 0.89 \pm 0.04$, $R^2_{\hat{V}^{\text{rest}}} = 0.76 \pm 0.09$ (outliers 14%)). This is consistent with our identifiability analysis. Per-neuron noise breaks the degenerate directions of $\mathbf{A}_i$, and is what enables the GNN to match the Known-ODE upper bound at $\sigma > 0$. The learned latent embedding $\mathbf{a}_i$ alone separates the 65 cell types, and combining it with $(\hat{\boldsymbol{\tau}}, \hat{\mathbf{V}}^{\text{rest}})$ and summary statistics of $\hat{\mathbf{W}}$ matches the clustering accuracy of the same procedure applied to ground-truth parameters (0.92 vs. 0.93 at $\sigma = 0.05$; Supp. Fig. 7 and Tab. 1).

Edge ablation as a mechanistic test. We propose an edge-ablation test to certify that a learned model is mechanistic, beyond merely predicting activity. For each of the three model-noise regimes ($\sigma \in \{0,\ 0.05,\ 0.5\}$), we randomly zero 50% of the 434,112 edges, regenerate 8,000 frames of held-out test activity with the perturbed connectome, and apply the same edge mask to the trained GNN's learned weights $\hat{\mathbf{W}}$ before running the autoregressive rollout. No retraining is performed, the test is whether $\hat{\mathbf{W}}$, $\hat{\boldsymbol{\tau}}$, $\hat{\mathbf{V}}^{\text{rest}}$, f_θ, g_ϕ extrapolate to a circuit with half its edges removed. The GNN passes this test at all model noise levels. The per-neuron Pearson correlation between the ablated rollout and the noise-free ablated simulation is $r \geq 0.99$ across all three noise levels (Supp. Fig. 8). The agreement holds for individual cell-type traces (Supp. Fig. 8a–c) and across the full voltage range (Supp. Fig. 8d–f). The high cross-condition agreement indicates that the learned f_θ and g_ϕ encode the local update rule rather than memorising trajectories, satisfying our mechanistic criterion and complementing the parameter-recovery results of Fig. 1.

Data degradation (measurement noise, partial data, unknown stimulus). We evaluate the GNN ($\sigma = 0.05$) on a grid of degradations expected in real experiments: measurement noise ($\gamma \in \{0.1, 0.2\}$), connectome errors as false positives ($+400\%$ random null edges, $5\times$ inflated graph) and false negatives (-20%, -50% edges removed), partial temporal sampling (1/5 frames), partial spatial sampling (10%, 20% neurons hidden), and unknown stimulus jointly recovered with a SIREN network (Appx. F.3 and Supp. Tab. 4). Activity prediction is robust across all conditions (one-step $r \geq 0.95$ except 1/5 frames at 0.86). Parameter recovery is most resilient to false-positive edges. The GNN correctly identifies the 80% of inflated edges as zero-weight and recovers the true $\hat{\mathbf{W}}$ at $R^2 = 0.96$ even at $5\times$ the true edge count. It degrades under -20% edge removal ($R^2_{\hat{W}} = 0.78$, rollout $r = 0.99$), but breaks down at -50% removal ($R^2_{\hat{W}} = 0.19$, rollout $r = 0.56$).

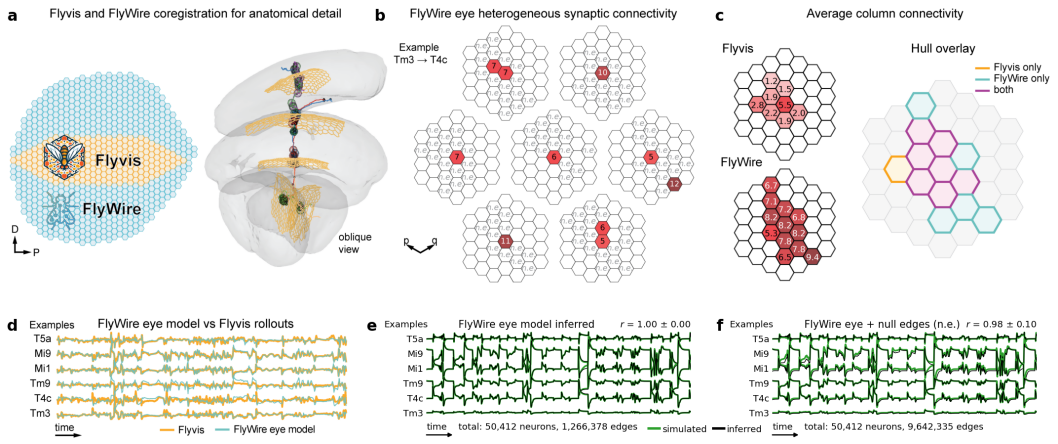


Figure 2: **Hybrid Flyvis–FlyWire connectome model inference.** **(a)** Hybrid model combining the Flyvis convolutional template (represented by orange disc, extent 8) with the FlyWire FAFB-v783 column lattice (teal disc) on the right hemisphere; 3D oblique view, right optic-lobe with representative registered visual neurons through lamina, medulla, lobula, and lobula plate neuropil meshes. Orange discs indicate "e8 hybrid" columns, scatter points indicate synapse positions, circles indicate 6 μm edge confusion radius around synapse clouds. **(b)** In contrast to Flyvis template connectivity (Supplementary Supp. Fig. 11), per-column FlyWire receptive-field connectivity for an example projection (Tm3 $\rightarrow$ T4c): Each mini-hexlattice is the spatial RF kernel of one T4c target column at the location $(p, q) \in \{(0,0),(0,1),(1,0),\cdots\}$ (extent 1), with input cells annotated by per-edge synapse count. Cells lacking FlyWire evidence but covered by a null edge (n.e.) are marked *n.e.* **(c)** Average column connectivity for the same Tm3 $\rightarrow$ T4c projection. *Flyvis*: per-column-averaged convolutional kernel from the original Flyvis template. *FlyWire*: per-(dp, dq) offset average of FlyWire synapse counts. *Hull overlay*: union of offset support, coloured by membership (orange = Flyvis only, teal = FlyWire only, magenta = both). **(d-f)** Rollout comparisons to 20,000 ms unseen DAVIS stimuli. **(d)** Rollouts of FlyWire eye model (teal) vs. Flyvis rollout (orange). **(e)** Rollouts of the general-form GNN ($g_\phi = \mathrm{MLP}(\mathbf{a}_i, \mathbf{a}_j, v_i, v_j)$, group lasso 25; black) vs. FlyWire eye simulations (green), fold cv00, with the Fisher-z pooled Pearson r over all 50,412 neurons and 7,207 frames. **(f)** The same GNN trained with 661% false-positive edges (FlyWire eye + n.e.) vs. FlyWire eye simulations.

When half the true edges are missing from the prior, the remaining edges cannot absorb their contribution. Measurement noise primarily corrupts $\hat{\mathbf{W}}$ ($R^2 = 0.64$ at $\gamma = 0.1$, 0.39 at $\gamma = 0.2$) while leaving rollout largely intact. Partial observation of frames or neurons preserves τ recovery ($R^2_\tau \geq 0.83$) but undermines $\hat{\mathbf{W}}$. As a first attempt to close this gap, we couple the GNN with an Instant-NGP coordinate network [Müller et al., 2022] that learns to fill in the unobserved neuron activity. Preliminary results recover part of the lost $\hat{\mathbf{W}}$ signal but do not help in parameter recovery (data not shown). When the input stimulus is itself hidden, we couple the GNN with a SIREN coordinate network [Sitzmann et al., 2020] that learns the photoreceptor drive jointly with the dynamics (Supp. Fig. 9 and Appx. F.3). On 5-fold CV (unknown-stimulus row of Supp. Tab. 4), the joint optimisation reconstructs a stimulus tensor of 1,736 photoreceptors $\times$ 64,000 frames (1.11×10^8 scalar values) and yields a rollout $r = 0.59 \pm 0.32$ on training-frame indices since the SIREN cannot extrapolate to coordinates unseen at training. $\hat{\boldsymbol{\tau}}$ ($R^2 = 0.93 \pm 0.04$) and $\hat{\mathbf{W}}$ ($R^2 = 0.81 \pm 0.18$) remain close to ground-truth values. Note that the stimulus is not truly unobserved. It is largely present in the activity of the 1,736 retinal photoreceptors (R1–R8 across 217 columns), which themselves also receive feedback from downstream neurons. The task is therefore to disentangle the feedforward stimulus drive from the recurrent feedback.

GNN recovers fly visual system simulation with real synaptic connectivity. While the convolutional assumption of the Flyvis model (Sect. 3) is a good approximation for the underlying biology, it discards per-column variation of connectivity that is present in the fly and available in the FlyWire whole-brain reconstructions. To test the counterfactual whether the GNN would recover a dynamical visual system with real biological synaptic wiring variability—or in other words, whether the GNN

Table 2: **GNN recovery on hybrid connectome variants** under connectivity uncertainty (null-edge augmentation); low model noise $\sigma = 0.05$, 5-fold CV (mean $\pm$ SD). Variant is either e8 hybrid (13,741 neurons) or FlyWire eye (50,412 neurons). Known-ODE and GNN rows are re-runs: the GNN is the general-form GNN, $g_\phi = \mathrm{MLP}(\mathbf{a}_i, \mathbf{a}_j, v_i, v_j)$ under a group lasso of 25; ML baselines are the published controls. Prediction metrics on noise-free held-out stimuli. Parameter recovery: $R^2_{\widehat{W}}$ by the template readout over the fitted non-zero edges, outlier-corrected; $R^2_{\hat{\tau}}$ and $R^2_{\hat{V}_{\mathrm{rest}}}$ over all neurons, outlier-corrected (Appendix I); (out. %): the percentage of outliers. Green: > 0.9.

					prediction		parameter recovery		
model	condition	neurons	edges	variant	one-step r	rollout r	$R^2_{\widehat{W}}$ (out. %)	$R^2_{\hat{\tau}}$ (out. %)	$R^2_{\hat{V}_{\mathrm{rest}}}$ (out. %)
Known ODE	het. RF	13 741	327 358	e8 hybrid	1.00±0	1.00±0	0.91±2E−4 (0%)	0.97±5E−4 (1%)	0.96±4E−4 (9%)
	+ n.e.	13 741	2 418 403	e8 hybrid	1.00±5E−4	1.00±0	0.90±8E−4 (0%)	0.97±5E−4 (0%)	0.90±8E−4 (7%)
larger visual field	het. RF	50 412	1 266 378	FlyWire eye	1.00±0	1.00±0	0.92±2E−4 (0%)	0.97±5E−4 (0%)	0.96±2E−4 (9%)
	+ n.e.	50 412	9 642 335	FlyWire eye	1.00±0	1.00±5E−4	0.89±9E−4 (0%)	0.97±2E−4 (1%)	0.86±6E−4 (8%)
GNN	het. RF	13 741	327 358	e8 hybrid	1.00±4E−4	1.00±6E−4	0.97±3E−3 (0%)	0.98±0.01 (0%)	0.88±0.01 (6%)
	+ n.e.	13 741	2 418 403	e8 hybrid	0.99±1E−3	0.99±3E−3	0.93±0.01 (0%)	0.94±0.01 (1%)	0.71±0.02 (20%)
larger visual field	het. RF	50 412	1 266 378	FlyWire eye	1.00±0	1.00±0	0.97±1E−3 (0%)	0.97±0.01 (0%)	0.91±0.01 (4%)
	+ n.e.	50 412	9 642 335	FlyWire eye	0.98±2E−3	0.98±3E−3	0.90±0.01 (0%)	0.94±0.01 (1%)	0.66±0.02 (28%)
MLP	het. RF	13 741	327 358	e8 hybrid	0.90±0.09	0.97±0.05	—	—	—
MLP *l.v.f.*	het. RF	50 412	1 266 378	FlyWire eye	0.86±0.10	0.96±0.06	—	—	—
EED	het. RF	13 741	327 358	e8 hybrid	0.88±0.12	0.97±0.05	—	—	—
EED *l.v.f.*	het. RF	50 412	1 266 378	FlyWire eye	0.83±0.14	0.95±0.06	—	—	—

inference capability was limited by the convolutional assumption—we constructed a hybrid connectome variant that augments the Flyvis graph with column-resolved FlyWire connectivity data, while remaining compatible with the trained Flyvis model parameters (Fig. 2a-d, Supp. Fig. 11, Sect. 3, and Appx. E).

We generated simulated data from the Flyvis ODE with low model noise ($\sigma = 0.05$) on FlyWire-based connectomes and trained the GNN and Known-ODE to recover the dynamical system.

On par with Known-ODE, the GNN trained on the e8 hybrid graph (327,358 edges across 13,741 neurons) achieved on unseen stimuli a one-step Pearson $r = 1.00 \pm 0.00$ and rollout $r = 1.00 \pm 0.00$, and inference remains as accurate as under the convolutional assumption (GNN low-noise in Tab. 1). Per-edge synaptic weight recovery was high ($R^2_{\hat{W}} = 0.94 \pm 0.04$), with time-constant recovery $R^2_{\hat{\tau}} = 0.99 \pm 0.01$ (0.1% outliers) and resting-potential recovery $R^2_{\hat{V}_{\mathrm{rest}}} = 0.87 \pm 0.03$ (6.9% outliers), confirming that the GNN can infer heterogeneous, column-specific connectivity accurately. Scaling to FlyWire eye (1,266,378 edges across 50,412 neurons, extent = 15) preserved this performance: one-step $r = 1.00 \pm 0.00$, rollout $r = 1.00 \pm 0.00$, $R^2_{\hat{W}} = 0.97 \pm 0.01$, $R^2_{\hat{\tau}} = 0.99 \pm 0.00$ (0.0% outliers), and $R^2_{\hat{V}_{\mathrm{rest}}} = 0.93 \pm 0.02$ (3.7% outliers) (Fig. 2e and Tab. 2).

GNN dynamical system inference overcomes false connectivity measurements. Reconstruction of brain wiring requires nanometer precision for accurate synaptic resolution of network connectivity. Sample preparation and imaging artifacts cause reconstruction errors leading to false measurements of connectivity.

Extending the degradation experiments in Sect. 4, we induced 'biologically plausible' false-positive connectivity into the graph: based on spatial proximity (within $\approx 6\,\mu$m, roughly one column in the fly visual system) of synapse clouds in the FlyWire anatomical coordinates, we estimated edges that were realistically confusable and included them in the connectivity measurement. The resulting connectivity graph augments the FlyWire-based connectome with these realistic candidate edges, increasing the edge count from 327,358 to 2,418,403 in e8 hybrid (13,741 neurons), and from 1,266,378 to 9,642,335 in FlyWire eye (50,412 neurons). This represents the worst case: every realistically confusable edge is included. In practice, the confused subset would be a random subsample of this set.

We again generated simulated data from the Flyvis ODE with neuronal noise ($\sigma = 0.05$) on this null-edge-augmented, FlyWire-based connectome and trained the GNN and known-ODE control to recover the dynamical system (Tab. 2). On the e8 hybrid graph, the GNN matches the known-ODE control on prediction (one-step $r = 1.00 \pm 0.00$, rollout $r = 1.00 \pm 0.01$) and recovers per-edge weights ($R^2_{\hat{W}} = 0.90 \pm 0.06$) and time-constants ($R^2_{\hat{\tau}} = 0.97 \pm 0.01$, 1.4% outliers) at near-clean

accuracy; resting-potential recovery degrades but remains informative ($R^2_{\hat{V}^{\text{rest}}} = 0.65 \pm 0.06$, 25.4% outliers). On the larger FlyWire eye, prediction stays accurate (Fig. 2f; one-step $r = 0.99 \pm 0.00$, rollout $r = 1.00 \pm 0.00$); weight recovery becomes more variable across folds ($R^2_{\hat{W}} = 0.83 \pm 0.16$), time-constants remain robust ($R^2_{\hat{\tau}} = 0.95 \pm 0.02$, 1.0% outliers), and the resting potential is the hardest parameter to identify ($R^2_{\hat{V}^{\text{rest}}} = 0.61 \pm 0.01$, 40.8% outliers). These results suggest that false-positive connectivity induced by measurement uncertainty and errors can be largely overcome with our system identification framework, with the residual difficulty concentrated in resting-potential identification at scale (Supp. Fig. 10).

5 Discussion

We asked how and when interpretable, parameter-resolved circuit models can be recovered from imperfect connectivity and activity, and addressed it on a 13.7K–50K neuron *Drosophila* visual-system simulator.

Interpretable GNN without prescribed dynamics. Our continuous-time GNN learns per-neuron update and per-synapse interaction functions jointly with effective weights, matching a Known-ODE oracle on weight, time-constant, and resting-potential recovery. Cell-type identity emerges unsupervised in the 2D latent embedding as a structural consequence of sharing f_θ and g_ϕ across neurons, and a SIREN extension additionally recovers unobserved photoreceptor drive. This bridges symbolic-search methods limited to small circuits [Lueckmann et al., 2026] and large-scale connectome-constrained models that commit to a fixed biophysical form [Mi et al., 2021, Lappalainen et al., 2024, Pospisil et al., 2024].

Identifiability across degradations. The connectome is most useful as a binary prior: tolerance to +400% random (Flyvis-217) and +661% spatially-plausible null edges (FlyWire) shows that knowing which pairs cannot connect matters more than precise weights, while removing 50% of true edges breaks recovery. For connectomics this shifts the quality target from synaptic-weight precision to edge-set completeness. Edge ablation provides a cheap mechanistic certificate—zeroing 50% of the connectome and masking $\hat{\mathbf{W}}$ accordingly yields rollout $r \geq 0.99$ without retraining—and is structurally inaccessible to predictors lacking a learned $\hat{\mathbf{W}}$; we propose it as a standard check for connectome-inverse claims.

Agentic HPO. Across six conditions, the loop converged on a single configuration in which the ℓ_1 prior on $\hat{\mathbf{W}}$ is the only load-bearing regularizer, consistent with the analytical degeneracy discussed in Appx. C. The agent's accumulated principles are themselves interpretable scientific output, not opaque numerical optima.

Perturbation richness, not architecture, is the binding constraint. ML baselines with 17–60$\times$ more parameters match one-step prediction but learn Jacobians inconsistent with the connectome (Supp. Fig. 18) and cannot support mechanistic interventions. Per-neuron noise lifts the columnar degeneracies and rescues recovery to oracle level, suggesting that perturbation-driven experiments (sparse optogenetics, voltage-clamp) are likely more informative than longer passive recordings. The analogy between intrinsic stochasticity and extrinsic perturbation remains to be tested.

6 Limitations

Several idealizations of the simulator do not transfer to real recordings. (1) Voltages are observed directly; calcium and voltage imaging are nonlinear, lower-bandwidth proxies. (2) Inference and simulation share $\Delta t = 20$ ms, so the piecewise-linear update is exact rather than approximate. (3) Conductance-based synapses are structurally unidentifiable from voltage alone—only the product of weight and driving force is constrained. (4) Static $\mathbf{W}, \tau, V^{rest}, and \mathbf{a}$ exclude neuromodulation and plasticity. (5) Flyvis is predominantly feedforward (cf. the $t_w = 16$ stimulus-only baseline at $r = 0.97$); stronger recurrent regimes are untested, though Allier et al. [2026] suggests they need not be harder. (6) Connectome-activity registration is perfect in silico; in practice it will be imperfect, partly mitigated by the demonstrated false-positive robustness.

Measurement noise on voltage traces is the principal open bottleneck: $\gamma = 0.1$ leaves rollouts intact but collapses $R^2(\hat{\mathbf{W}})$ from 0.99 to 0.63. Promising directions are longer rollouts to average

noise, likelihood-based losses matched to the noise model, and joint denoising via implicit neural representations as we did for the unknown stimulus (Appx. F.3). Compute scales steeply with graph size (Supp. Tab. 9), but the shared f_θ, g_ϕ make the model amenable to subgraph mini-batching, which we have not yet exploited.

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A Supplementary Tables

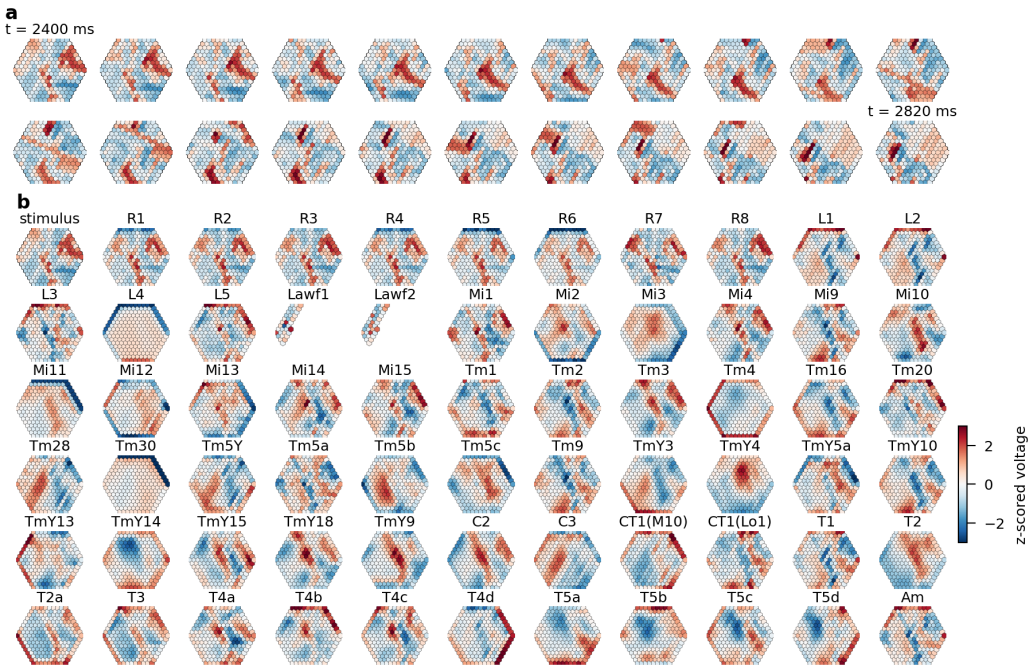
Supplementary Table 3: **Coregistration of Flyvis and FlyWire (FAFB-v783)**. "FlyWire eye" first tiles the Flyvis template (65 cell types, 2,355 edge templates) across every right-optic-lobe column reconstructed in FlyWire; "e8 hybrid" is the central hex-lattice crop (radius 8). The hybrid graph keeps edges from FlyWire where the postsynaptic neuron's receptive field was reconstructed in FlyWire and falls back to the Flyvis template edges elsewhere. The hybrid graph with uncertainty adds null edges (n.e.) based on spatially proximal synaptic connectivity.

	FlyWire eye	e8 hybrid (central crop)
Cell types	65	65
Nodes (type × column)	50,412	13,741
Flyvis-template edges	1,650,671	434,112
Edge templates	2,355	2,355
Registered nodes (assigned a FlyWire `root_id`)	32,698 (64.9%)	9,344 (68.0%)
Edges with both endpoints registered	800,489 (48.5%)	230,905 (53.2%)
Templates with ≥ 1 registered instance	2,203 (93.5%)	2,202 (93.5%)
Edges with FlyWire synapse coordinates	365,693 (22.2%)	106,370 (24.5%)
Total FlyWire synapses matched	3,800,254	1,174,678
Hybrid graph (het. RF)	1,266,378	327,358
Hybrid graph + null edges (n.e.; ≈6 µm, $n_{\mathrm{syn}} = 0$)	9,642,335	2,418,403

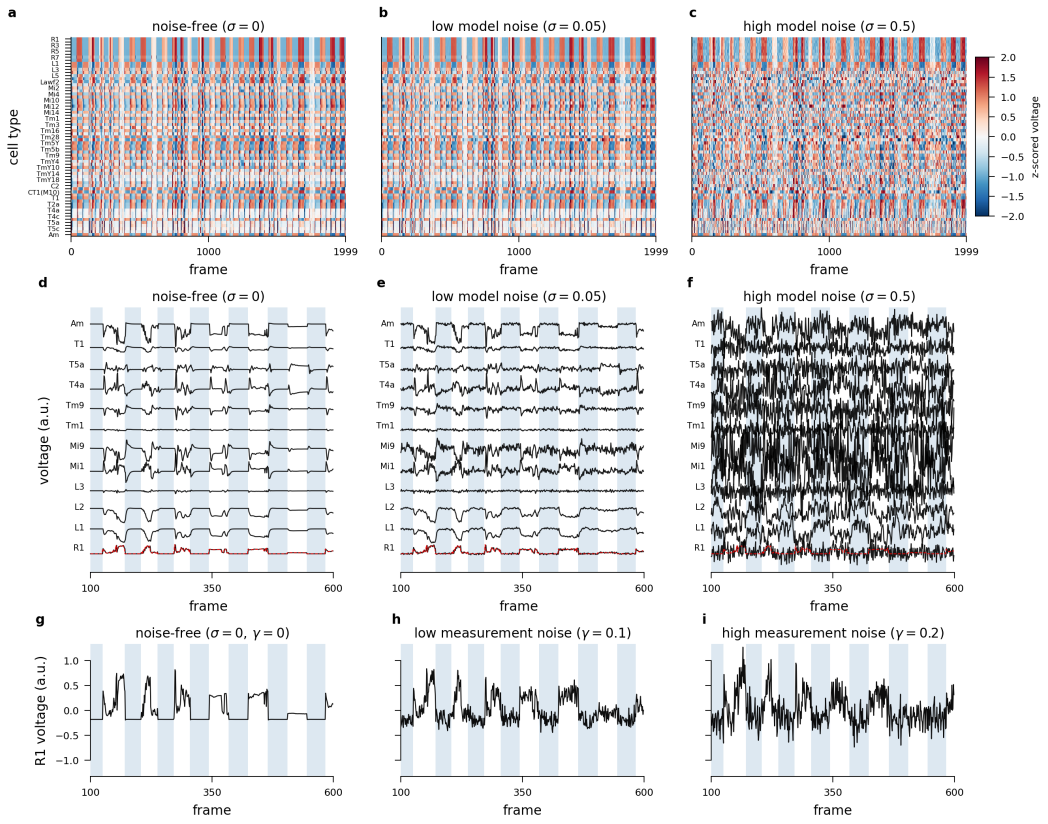
Supplementary Table 4: **GNN evaluation across degraded Flyvis-217**; model noise $\sigma = 0.05$, 5-fold CV (mean ± SD) with variable measurement noise (γ), unknown stimulus, added/removed edges, sub-sampled frames, and hidden neurons. Prediction metrics on noise-free held-out stimuli. Parameter recovery: $R^2_{\hat{W}}$ over the fitted edges, outlier-corrected; $R^2_{\hat{\tau}}$ and $R^2_{\hat{V}_{\mathrm{rest}}}$ over all 13,741 neurons, outlier-corrected (Appendix I); GMM clustering accuracy over 65 cell types. Green: > 0.9. Orange: < 0.3. (Among 10 trained folds the rollout from the initial voltages diverged ($r < 0.5$) in 3 for the unknown stimulus (cv03, cv04, cv06) and in 2 for +400% null edges (cv02, cv08); in these two rows each outlier fold of cv00–cv04 is replaced by the next fold that is not one.)

	measurement		prediction		parameter recovery			
condition	noise γ	edges	one-step r	rollout r	$R^2_{\hat{W}}$ (out.%)	$R^2_{\hat{\tau}}$ (out.%)	$R^2_{\hat{V}_{\mathrm{rest}}}$ (out.%)	cluster. acc.
low model noise	0	434 112	1.00±0	1.00±4E−4	0.98±2E−3 (0%)	0.98±4E−3 (0%)	0.89±0.02 (3%)	0.90±0.01
low meas. noise, 20-step recurrent	0.1	434 112	0.98±2E−3	0.99±7E−4	0.77±0.01 (0%)	0.53±0.05 (5%)	0.66±0.03 (25%)	0.85±0.01
mid meas. noise, 20-step recurrent	0.2	434 112	0.95±3E−3	0.98±1E−3	0.61±0.01 (0%)	0.20±0.03 (10%)	0.58±0.02 (37%)	0.81±0.01
unknown stimulus	0	434 112	1.00±5E−4	0.71±0.06	0.91±0.03 (0%)	0.85±0.08 (2%)	0.77±0.06 (24%)	0.87±0.03
+400% null edges	0	2 170 560	1.00±5E−4	1.00±7E−4	0.96±0.02 (0%)	0.96±0.01 (1%)	0.47±0.05 (16%)	0.85±0.02
−20% edges removed	0	347 000	0.99±5E−4	0.99±10E−4	0.87±0.01 (0%)	0.97±0.01 (0%)	0.73±0.02 (15%)	0.70±0.02
−50% edges removed	0	217 056	0.97±2E−3	0.93±3E−3	0.43±0.01 (0%)	0.95±0.01 (0%)	0.54±0.02 (36%)	0.58±0.02
1/5 frames, rollout horizon 21	0	434 112	1.00±5E−4	1.00±4E−4	0.95±0.01 (0%)	0.93±0.02 (0%)	0.84±0.01 (10%)	0.87±0.01
10% hidden	0	434 112	0.99±4E−4	0.88±2E−3	0.83±2E−3 (0%)	0.96±0.01 (3%)	0.82±0.02 (14%)	0.71±0.01
20% hidden	0	434 112	0.98±2E−3	0.88±2E−3	0.65±0.02 (0%)	0.92±0.02 (5%)	0.72±0.02 (27%)	0.55±0.02

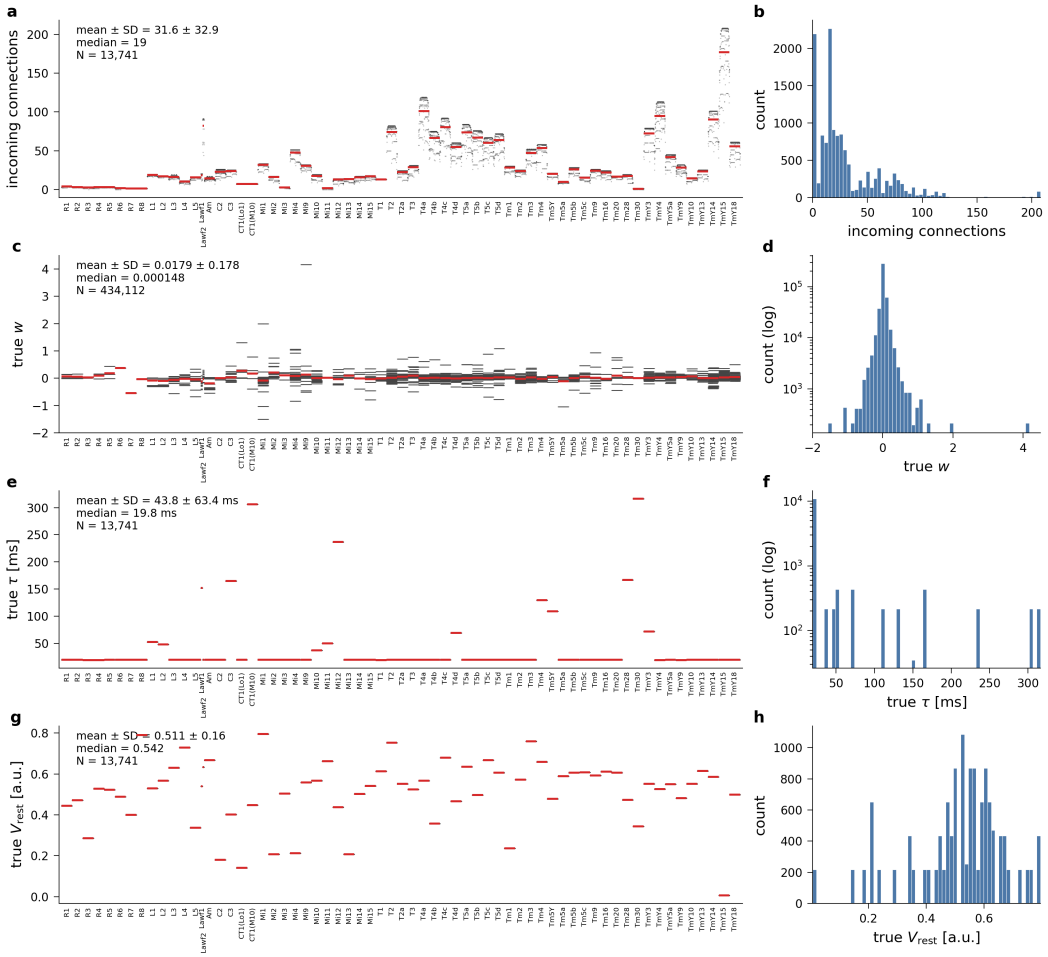
B Supplementary Figures



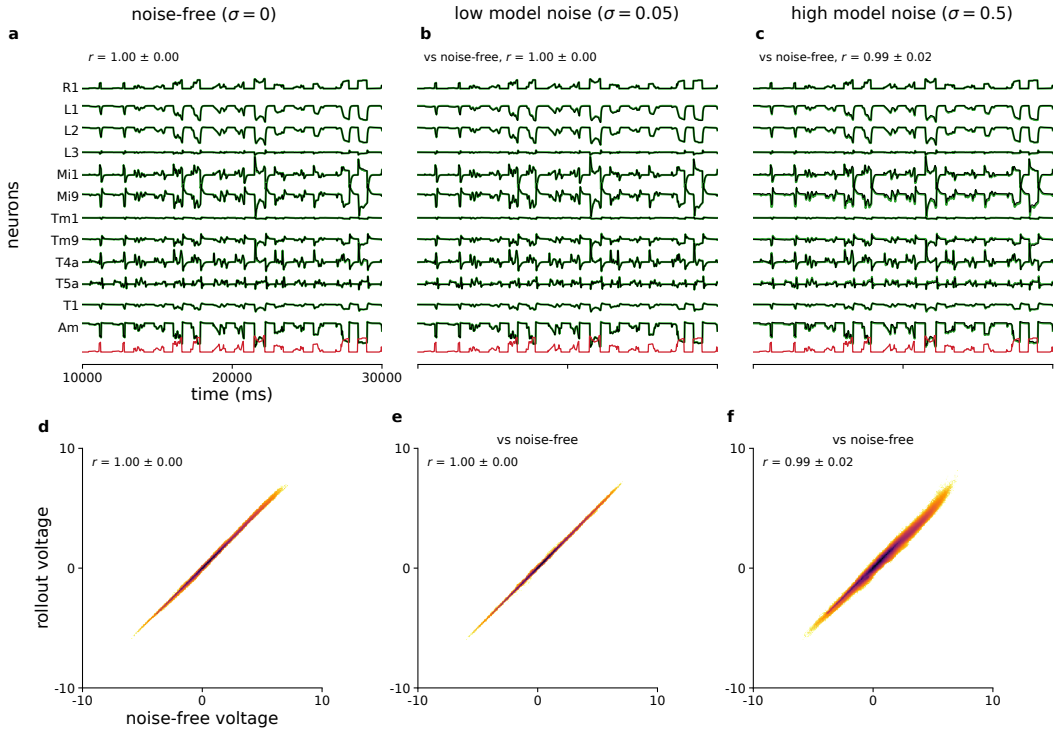
Supplementary Figure 3: **Visual input and per-cell-type activity** on the retinotopic hexagonal lattice (central 217-column Flyvis model, noise-free, 13,741 neurons, 1,736 photoreceptors, 434,112 synaptic connections, integration step $\Delta t = 20$ ms). **(a)** Consecutive frames of the visual stimulus ($t = 2400$–2820 ms, one panel every 20 ms) projected on the hex lattice of photoreceptors. Each frame is z-scored across photoreceptors so that local contrast pattern is visible. **(b)** Snapshot of the voltage at $t = 2400$ ms for the stimulus (top-left) and the 65 cell types. Values are z-scored *within* each cell type, so every hexagon panel shares the same color scale (shown on the right). Red (positive) and blue (negative) indicate deviations from the type's mean voltage at that instant; white is at the mean.



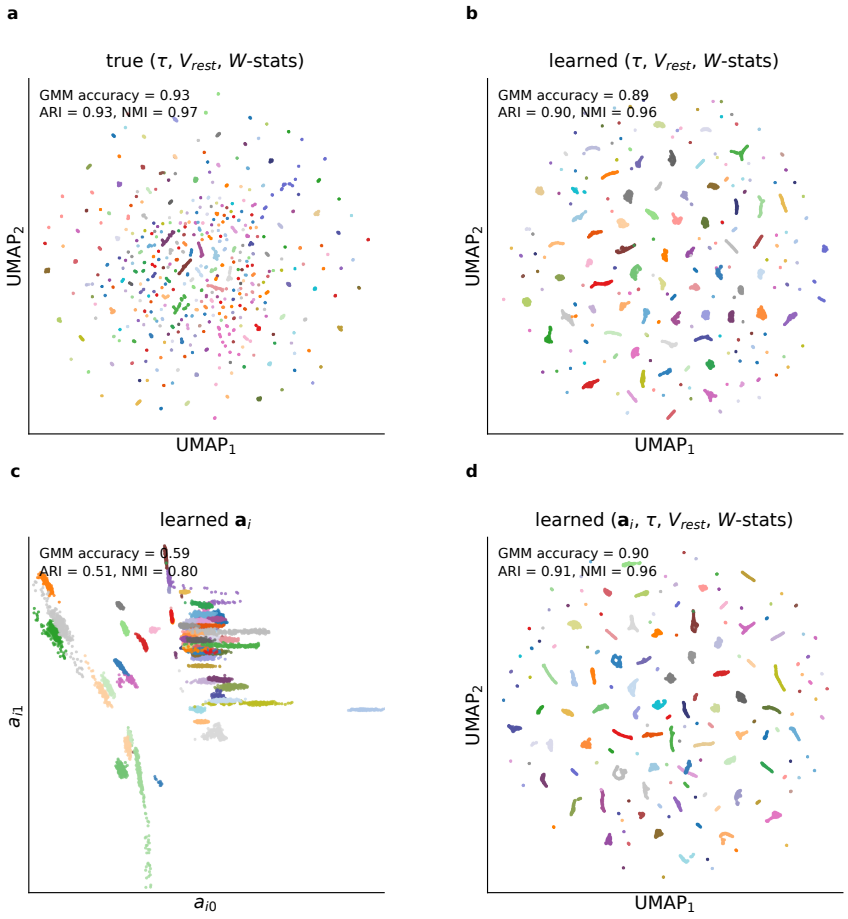
Supplementary Figure 4: **Simulated central** 217**-column Flyvis activity across noise regimes.** Voltage traces $v_i(t)$ are generated by Eq. (1); $\xi_i(t) \sim \mathcal{N}(0,1)$ is an independent Gaussian noise scaled by σ (model noise), perturbing the dynamical trajectory of $v_i(t)$. We additionally apply a measurement noise directly to $v_i(t)$, Gaussian noise scaled by γ. All simulations use 50% null stimulus; blue spans mark null frames. (**a**–**c**) Type-mean voltage heatmap, z-scored per type; columns are noise-free ($\sigma = 0$), low model noise ($\sigma = 0.05$), high model noise ($\sigma = 0.5$). (**d**–**f**) Voltage traces of 12 representative cell types over a 500-frame window; same three model-noise columns. Red line: stimulus. (**g**–**i**) Zoom on a single R1 photoreceptor at three measurement-noise levels ($\gamma = 0, 0.1, 0.2$).



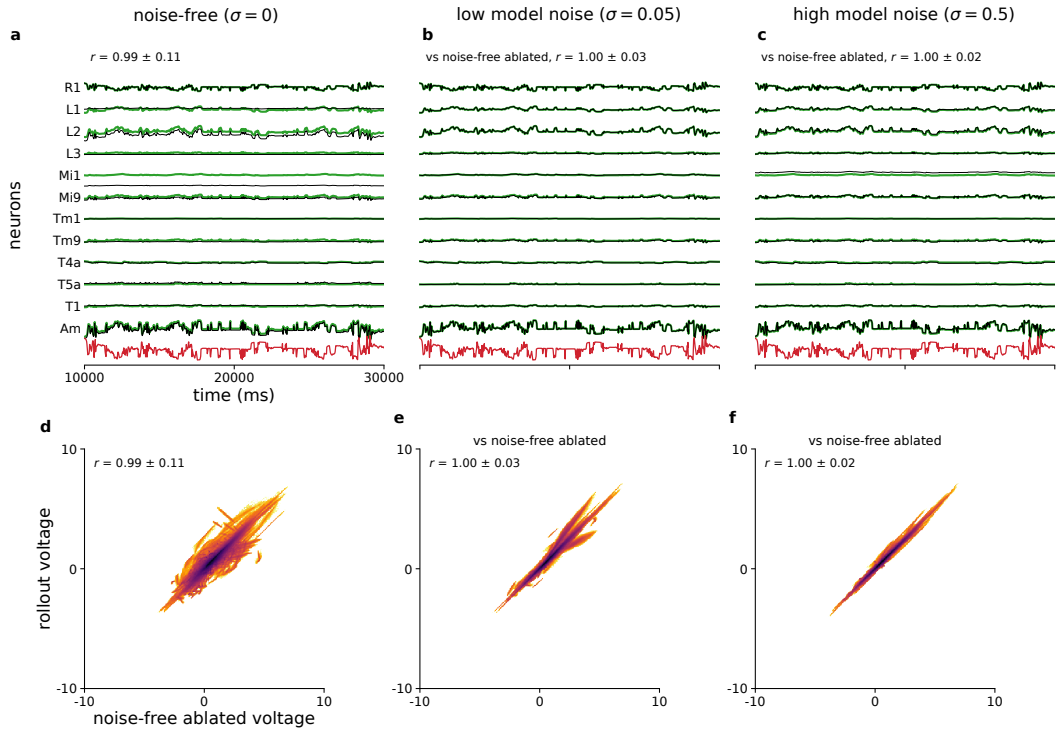
Supplementary Figure 5: **Ground-truth circuit-parameter distributions of the central** 217**-column Flyvis model** (13,741 neurons, 65 cell types, 434,112 synaptic edges). **(a)** Number of incoming synaptic connections per neuron (one dot per neuron, sorted by cell type; red horizontal lines mark the per-type mean). **(b)** Marginal histogram of the same quantity. **(c)** True synaptic weight W_{ij}, one dot per edge ($y = w_{j\to i}$ plotted at $x = i$). **(d)** Per-edge weight histogram (log y); sharply peaked at 0 with a thin tail to $w \approx 4$. **(e)** True membrane time constant τ_i. **(f)** τ histogram (log y); bimodal with $\sim 79\%$ of neurons at $\tau \approx 25\,\mathrm{ms}$ and a sparse tail out to $\sim 320\,\mathrm{ms}$. **(g)** True resting potential V_i^{rest}. **(h)** V^{rest} histogram; concentrated in 0.4–0.7. Top-left of each left-column panel: mean, median, and sample size.



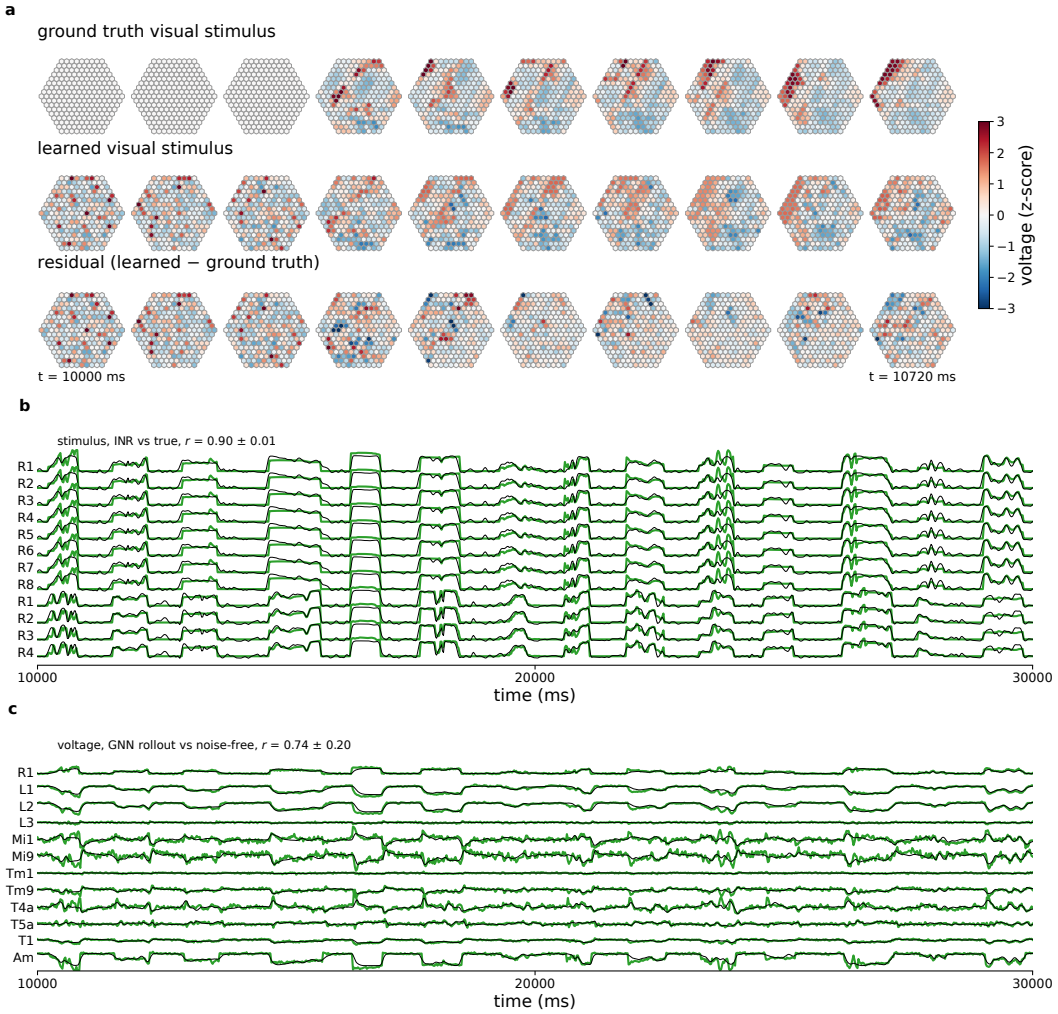
Supplementary Figure 6: **Rollout prediction.** General-form GNN (group lasso 25) evaluated on central 217-column Flyvis data under three model noise regimes ($\sigma = 0$, $\sigma = 0.05$, $\sigma = 0.5$; fold cv00), tested on held-out stimuli, and compared against the noise-free Flyvis simulation of the same stimuli. Top row (**a**, **b**, **c**): GNN rollout traces for 12 representative cell types over a 20 s window (1,000 frames at $\Delta t = 20$ ms). Green: noise-free ground-truth voltage; black: GNN rollout prediction; red: the visual input of the photoreceptor row. Bottom row (**d**, **e**, **f**): rollout voltage against noise-free ground-truth voltage, pooled over all (neuron, frame) pairs of the 7,207-frame rollout (the fold's held-out stimuli) as a log-density image. Pearson r: per-neuron, Fisher-z pooled, on the full rollout.



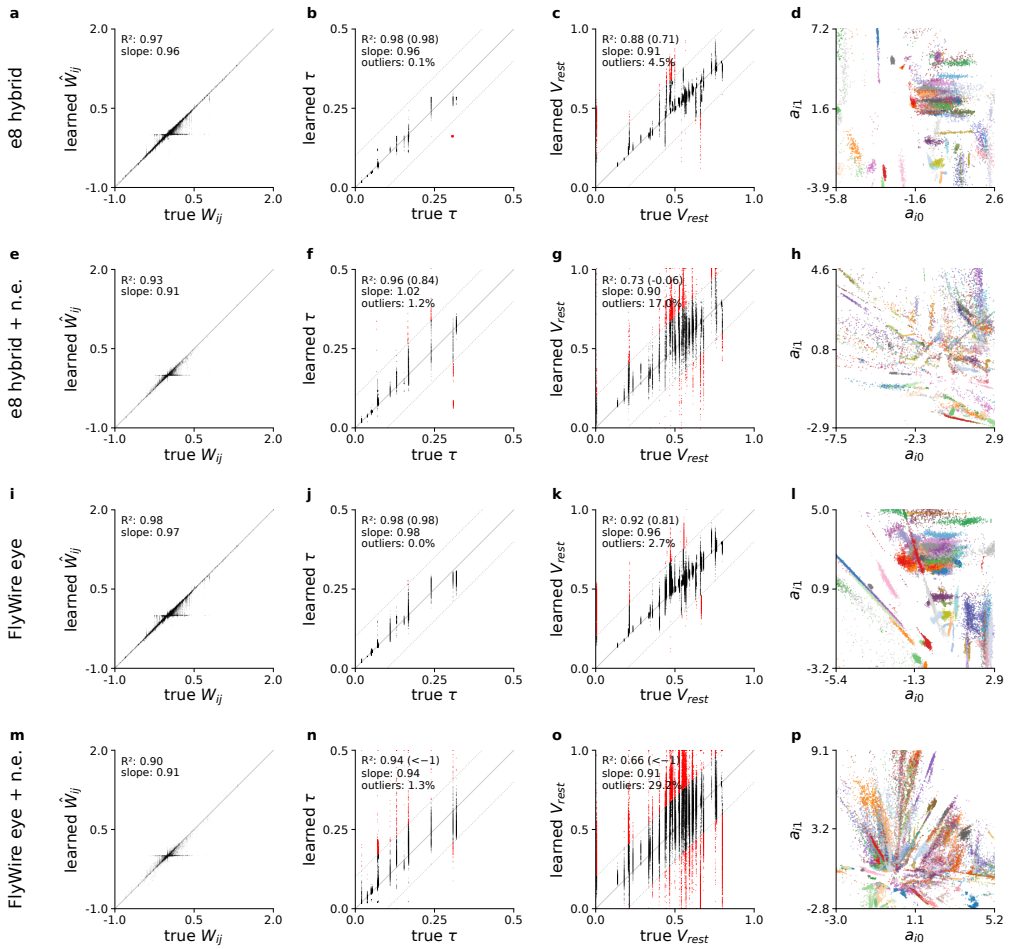
Supplementary Figure 7: **Cell-type clusterability across feature spaces** of the general-form GNN (group lasso 25, $\sigma = 0.05$, fold cv00). Each panel scatters the 13,741 neurons of the central 217-column Flyvis model coloured by ground-truth cell type (65 classes). Gaussian Mixture Model (GMM, 100 components) accuracy, Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI), quoted top-left of each panel, are the analysis pipeline's own scores of each feature stack (the stack standardised per feature); UMAP is shown for visualisation only when the feature space is > 2D. (**a**) Ground-truth (τ, V_{rest}) plus 8 connectivity statistics of the true $\mathbf{W}$ (mean, SD, minimum and maximum of the incoming and of the outgoing weights; 10D). (**b**) The same from the learned $\hat{\tau}$, $\hat{V}_{\text{rest}}$ and $\hat{\mathbf{W}}$ (10D). (**c**) Learned 2D node embedding $\mathbf{a}_i$ shown directly (no projection). (**d**) $\mathbf{a}_i$ concatenated with the learned $(\hat{\tau}, \hat{V}_{\text{rest}}, \hat{\mathbf{W}}\text{-stats})$ (12D): the stack of the clustering accuracy reported in Tab. 1.



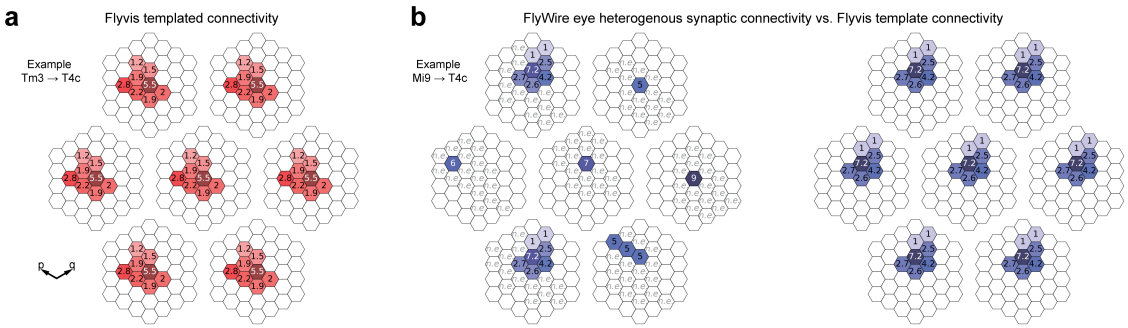
Supplementary Figure 8: **Rollout prediction under 50% edge ablation.** General-form GNN (group lasso 25; fold cv00) evaluated on central 217-column Flyvis data under three model noise regimes (noise-free $\sigma = 0$, $\sigma = 0.05$, $\sigma = 0.5$), tested on held-out stimuli, and compared against the noise-free ablated Flyvis simulation. For each test the simulator regenerates voltage traces with 50% of the synaptic edges zeroed, and the same edge mask is applied to the trained GNN's learned weights $\hat{\mathbf{W}}$ before rollout (no retraining). Top row (**a**, **b**, **c**): GNN rollout traces for 12 representative cell types over a 20 s window (1,000 frames at $\Delta t = 20$ ms). Green: ablated ground-truth voltage; black: GNN rollout prediction; red: the visual input of the photoreceptor row. Bottom row (**d**, **e**, **f**): rollout voltage against the noise-free ablated voltage, pooled over all $(\text{neuron}, \text{frame})$ pairs of the 7,999-frame rollout as a log-density image. Pearson r: per-neuron, Fisher-z pooled, on the full rollout.



Supplementary Figure 9: **Joint GNN+INR (visual SIREN)**: the general-form GNN (group lasso 25, one-step training) evaluated on central 217-column Flyvis data with low model noise ($\sigma = 0.05$), 64 000 training frames (fold cv00); rollout evaluated on the first 7,999 training frames since the SIREN cannot extrapolate to test-time indices. The SIREN output is shown with its mean and SD matched to the true stimulus, sign from the correlation (its raw output is defined up to the sign and scale the GNN's input weights absorb; the tester's least-squares correction would compress the learned range by the factor r). **(a)** INR-reconstructed visual stimulus on the R1 photoreceptor lattice (217 columns), ten frames spaced by 80 ms from 10,000 ms, z-scored per frame; top: ground truth; middle: INR; bottom: residual. **(b)** INR stimulus (black) against the true one (green) for 12 photoreceptors (R1–R8 of one column, R1–R4 of another) over a 20 s window (1,000 frames at $\Delta t = 20$ ms); the header gives the per-photoreceptor Pearson r over the 7,999 frames, Fisher-z pooled over the 1,736 photoreceptors (mean ± SD). The SIREN's output enters the GNN squared, so it cannot be negative, and f_θ is free to read it with either sign; this run learned the inverted code (bright pixels → small output), so its zero floor becomes a ceiling at 0.73 of the true range: the 7% of photoreceptor–frames brighter than it all receive the same input, visible as the flattened bright plateaus. The sign is set at random by the training (4 of the 5 folds learned it inverted). **(c)** GNN voltage rollout (black) against the noise-free ground truth (green) for 12 representative cell types over the same window; the header gives the per-neuron Pearson r, Fisher-z pooled over all 13,741 neurons (mean ± SD).



Supplementary Figure 10: **Circuit parameter extraction from the general-form GNN** (group lasso 25) **on the four FlyWire connectome variants** with low model noise ($\sigma = 0.05$; fold cv00): e8 hybrid, e8 hybrid with proximal null edges (n.e.), FlyWire eye, and FlyWire eye with proximal null edges, one row each. **(a, e, i, m)** Learned against true synaptic weight $\hat{W}_{ij}$ over the edges fitted by the template readout (95%, 95%, 95% and 94% of the edges). **(b, f, j, n)** Time constants τ_i and **(c, g, k, o)** resting potentials V_i^{rest} of the template readout, outliers beyond the dotted ± 0.1 s / ± 0.2 bands in red; R^2 (inlier, all-neuron in parentheses), slope and outlier fraction as in Tab. 2. **(d, h, l, p)** Learned latent embeddings $\mathbf{a}_i \in \mathbb{R}^2$ of all neurons, coloured by ground-truth cell type.



Supplementary Figure 11: **Hybrid model heterogenous connectivity** examples vs. Flyvis templated connectivity. Each lattice represents the columnar input grid to a postsynaptic T4c at the location $(p, q) \in \{(0,0),(0,1),(1,0),\cdots\}$. Individual hexagons show synapse counts from presynaptic neurons of the source type at column locations in the same p, q-grid to the respective T4c cell. **(a)** For illustration, in Flyvis, the connectivity is based on the same template from a single-column reconstruction that is tiled across columns to compile a full eye model. In contrast, as seen in **(b)**, our Hybrid model introduces real columnar connectivity variability using the FlyWire (FAFB-v783) data where available.

C Degeneracy of the inverse problem

Recall that the Flyvis stochastic dynamical system in Eq. (1), in the absence of noise ($\sigma = 0$), takes the form:

$$\tau_i \frac{dv_i(t)}{dt} = -v_i(t) + V_i^{\text{rest}} + \sum_{j \in \mathcal{N}_i} W_{ij}\, \text{ReLU}\big(v_j(t)\big) + I_i(t) \tag{5}$$

Our goal is to infer the circuit parameters $(\tau_i, V_i^{\text{rest}}, W_{ij})$ given a solution of the ODE $\tilde{v}_i(t)$ and the stimulus $I_i(t)$ sampled at T discrete points $t = 0, \cdots T-1$. The ODE parameters can be decomposed into a separate set for each neuron i, namely, $\theta_{i\alpha} := (\tau_i, V_i^{\text{rest}}, W_{ij})$ is a vector of length $2 + d_i$ indexed by α, where $\mathcal{N}_i$ is the set of presynaptic partners of neuron i and $d_i := |\mathcal{N}_i|$ is the in-degree. We assume that the solution $\tilde{v}_i(t)$ was generated by a parameter set θ and solving the inverse problem gives us an inferred set $\hat{\theta}$. We show here that, in the absence of noise, the parameter determination is ill-posed and that there are *continuous* families of learned $\hat{\theta}$ consistent with the given $\tilde{v}_i$ and I_i.

Supp. Eq. 5 can be rewritten, for the purposes of parameter inference

$$-\tau_i \dot{\tilde{v}}_i + V_i^{\text{rest}} + \sum_{j \in \mathcal{N}_i} W_{ij} h_j(t) = \tilde{v}_i(t) - I_i(t) \tag{6}$$

where $h_j(t) := \text{ReLU}(\tilde{v}_j(t))$ are the pre-synaptic activities. Observe that while the ODE is non-linear in the state variable v it is *linear* in the circuit parameters $\theta_{i\alpha}$ defined above. So, we can rewrite Supp. Eq. 6 as a separate linear system to solve for each neuron i,

$$\mathbf{A}_i\, \boldsymbol{\theta}_i = \mathbf{b}_i, \tag{7}$$

in terms of the design matrix $\mathbf{A}_i$ of dimension $T \times (2 + d_i)$, and the vectors $\boldsymbol{\theta}_i$ of dimension $2 + d_i$ and $\mathbf{b}_i$ of dimension T:

$$\mathbf{A}_i := \begin{pmatrix} -\dot{\tilde{v}}_i(0) & 1 & h_{j_1}(0) & h_{j_2}(0) & \cdots \\ -\dot{\tilde{v}}_i(1) & 1 & h_{j_1}(1) & h_{j_2}(1) & \cdots \\ \vdots & \vdots & \vdots & \vdots & \end{pmatrix}, \quad \boldsymbol{\theta}_i := \begin{pmatrix} \tau_i \\ V_i^{\text{rest}} \\ W_{ij_1} \\ W_{ij_2} \\ \vdots \end{pmatrix}, \quad \mathbf{b}_i := \begin{pmatrix} \tilde{v}_i(0) - I_i(0) \\ \tilde{v}_i(1) - I_i(1) \\ \vdots \end{pmatrix} \tag{8}$$

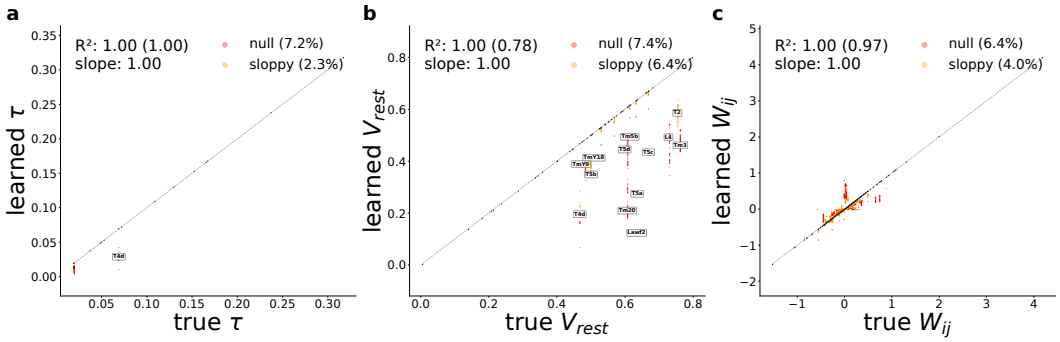
where $j_1, j_2, \ldots$ are the pre-synaptic neurons in $\mathcal{N}_i$. We emphasise that this linear system is not a linearisation or approximation but a property of the voltage-based firing-rate models used in Flyvis [Lappalainen et al., 2024].

The linear system Supp. Eq. 7 can be solved to infer the simulation parameters $\theta_{i\alpha}$. If the column space is not full rank the system admits a continuous degeneracy defined by the kernel $\ker(\mathbf{A}_i)$. In particular, if there exists a vector $\boldsymbol{\eta}_i$ satisfying $\mathbf{A}_i\, \boldsymbol{\eta}_i = \mathbf{0}$ for every i, then the family of parameters $\boldsymbol{\theta}_i' = \boldsymbol{\theta}_i + \lambda\, \boldsymbol{\eta}_i$ are solutions to the inverse problem for all $\lambda \in \mathbb{R}$. We solve the linear system using least squares, using float64 arithmetic, column normalization of $\mathbf{A}_i$. We compute the singular value decomposition of the $(2 + d_i) \times (2 + d_i)$ Gram matrix $\mathbf{A}_i^T \mathbf{A}_i$ and use the ratio λ/λ_{max} of singular values to define null (ratio $< 64000 \times 10^{-22}$) and sloppy (ratio $< 64000 \times 10^{-12}$) directions in the parameter space [Gutenkunst et al., 2007]. The results, shown in Supp. Fig. 12, demonstrate that with the DAVIS video stimulus, inference of τ, V^{rest}, W parameters from noise-free neural activity is indeed ill-posed.

If the network structure and stimulus are such that a neuron i is in a resting state throughout the experiment, then we have $\tilde{v}_i(t) = V_i^{\text{rest}}$ and $\dot{\tilde{v}}_i(t) = 0$ for all observed t. In this case, the first column of $\mathbf{A}_i$ vanishes and τ_i is completely unconstrained by the data. We have verified this to be the case for all neurons in Supp. Fig. 12A. Similarly, if a neuron j is active but $v_j(t) < 0$ for all time then the $h_j(t)$ column of the matrix $\mathbf{A}_i$ is identically zero, causing a null direction along the corresponding $\hat{W}_{ij}$. There are no degeneracies *purely* along the V^{rest} direction because the corresponding column in $\mathbf{A}_i$ is identically 1. However, if a pre-synaptic neuron j is always at rest $v_j(t) = V_j^{\text{rest}}$, then the following transformation leaves the equations invariant:

$$\hat{W}_{ij} \to \hat{W}_{ij} + x, \; \hat{V}_i^{\text{rest}} \to \hat{V}_i^{\text{rest}} - V_j^{\text{rest}} x \tag{9}$$

The examples above help explain the presence of null or sloppy directions for each of the parameters in Supp. Fig. 12 despite a biological realistic connectivity and 64,000 frames of diverse video stimuli.



Supplementary Figure 12: **Per-neuron least-squares recovery** of the ground-truth ODE parameters on noise-free Flyvis data. Each panel scatters the recovered value against the ground truth; the dashed line is $y = x$. **(A)** membrane time constant τ_i; **(B)** resting voltage V_i^{rest}; **(C)** synaptic weight W_{ij}. Black points are well-identified; orange ("sloppy") and red ("null") points lie in poorly-conditioned directions of the per-neuron Gram matrix and are flagged as degenerate by the eigendecomposition. We report R^2 over the well-identified subset (with the all-points R^2 in parentheses) and the regression slope on the same subset. Fractions in the legend indicate the fraction of parameter dimensions that have non-zero inner product with a null or sloppy direction; in panels A, B it is the fraction of neurons, and in panel C it is the fraction of edges.

Highly discordant weights can generate similar rollouts

Above, we demonstrated that the inverse problem of determining the circuit parameters with perfect knowledge of the ODE and the connectivity edges between neuron pairs is ill-posed. Here, we show that due to the symmetries of the connectome, large perturbations of the weight matrices can leave the neural activity traces approximately invariant.

We fix the values of V^{rest} and τ and rearrange Supp. Eq. 6 yielding

$$\sum_{j \in \mathcal{N}_i} W_{ij} \operatorname{ReLU}\big(v_j(t)\big) = \tau_i \frac{dv_i(t)}{dt} + v_i(t) - V_i^{\text{rest}} - I_i(t). \tag{10}$$

Redefining the residual term $b_i(t) = \tau_i \frac{dv_i(t)}{dt} + v_i(t) - V_i^{\text{rest}} - I_i(t)$, we have

$$\mathbf{H}_i \, \mathbf{w}_i = \mathbf{b}_i, \tag{11}$$

where $\mathbf{H}_i \in \mathbb{R}^{T \times d_i}$ is the presynaptic activity matrix, d_i is the in-degree of neuron i, $\mathbf{w}_i \in \mathbb{R}^{d_i}$ the incoming weight vector, and $\mathbf{b}_i \in \mathbb{R}^T$ the stacked residuals. This holds for each postsynaptic neuron $i = 1, \ldots, N$, where $N = 13{,}697$ and d_i ranges from 1 to 208. The null space $\ker(\mathbf{H}_i) = \{\boldsymbol{\delta} \in \mathbb{R}^{d_i} : \mathbf{H}_i \boldsymbol{\delta} = \mathbf{0}\}$ characterizes the degeneracy: any $\mathbf{w}_i + \boldsymbol{\delta}$ with $\boldsymbol{\delta} \in \ker(\mathbf{H}_i)$ satisfies Eq. (11) equally and produces identical dynamics. In the following, we characterize the dynamical equivalence class of $\mathbf{W}$, the set of all weight matrices that reproduce the observed trajectories.

The columnar organization of the connectome directly implies the existence of $\ker(\mathbf{H}_i)$. The Flyvis circuit spans 217 retinotopic columns, and each postsynaptic neuron i typically receives input from $k_{i\alpha} > 1$ neurons of the same cell type α across columns. A moving stimulus crosses the eye so that column c sees what column $c - 1$ saw moments earlier: same-type neurons produce time-shifted copies of the same signal, giving the corresponding $k_{i\alpha}$ columns of $\mathbf{H}_i$ rank ≈ 1 instead of $k_{i\alpha}$. Any sum-zero perturbation $\sum_{m=1}^{k_{i\alpha}} \delta_{j_m} = 0$ on those weights leaves $\mathbf{H}_i \mathbf{w}_i$ unchanged, contributing $k_{i\alpha} - 1$ free directions to $\ker(\mathbf{H}_i)$. Summing over all neurons and cell types:

$$\dim \ker(\mathbf{H}) = \sum_{i=1}^{N} \sum_{\alpha:\, k_{i\alpha} > 1} (k_{i\alpha} - 1) = 308{,}160 \quad (71\% \text{ of edges}). \tag{12}$$

To verify Eq. (12), we generated one single-type variant per degenerate cell type: a perturbed matrix $\mathbf{W}' = \mathbf{W} + \boldsymbol{\Delta}$, where $\boldsymbol{\Delta}$ applies sum-zero perturbations at maximum amplitude ($8\times$ mean group weight) within the degenerate groups of that type. Of the 65 Flyvis cell types, 11 have no degenerate

groups and 2 (Lawf2, Tm5b) have no outgoing edges; the remaining 52 types each yield one variant. For each we computed $R^2_{\mathbf{W}}$ (weight-space agreement with GT) and rollout Pearson r (ODE trajectory agreement over 1,000 frames). $R^2_{\mathbf{W}}$ ranges from 0.28 to 1.0 while mean rollout Pearson r remains above 0.94 (min: TmY10). Perturbed weights differ substantially from the ground truth yet produce nearly identical dynamics (Supp. Tab. 5). Further, we looked for the minimal-edges $\mathbf{W}$ by collapsing each degenerate (i, α) group of $k_{i\alpha}$ edges onto a single representative edge $e^* = \arg\max_e |W^*_e|$, zeroing the remaining $k_{i\alpha} - 1$ edges. We tested two variants: sum-preserving ($W_{e^*} = \sum_e W^*_e$, exact null-space member) and calibrated (W_{e^*} minimising the per-group dynamics residual via one least-squares scalar fit). Both zero 308,160 edges (71% of 434,112). The sum-preserving variant achieves mean rollout Pearson $r = 0.95$ with $R^2_{\mathbf{W}} = 0.10$. The calibrated variant achieves mean rollout Pearson $r = 0.98$ with $R^2_{\mathbf{W}} = 0.39$.

Cell Type	$R^2_{\mathbf{W}}$	rollout r	Cell Type	$R^2_{\mathbf{W}}$	rollout r	Cell Type	$R^2_{\mathbf{W}}$	rollout r
R2	0.94	1.00	R3	0.97	1.00	R4	0.74	1.00
R5	0.99	1.00	R6	1.00	1.00	R7	1.00	1.00
R8	0.90	1.00	L1	0.73	0.98	L2	0.62	1.00
L4	1.00	1.00	L5	0.73	0.99	Lawf1	1.00	1.00
Am	0.95	1.00	C2	0.93	1.00	C3	1.00	1.00
CT1(Lo1)	1.00	1.00	CT1(M10)	0.88	1.00	Mi2	0.91	0.98
Mi3	0.94	1.00	Mi14	0.96	1.00	Mi15	0.87	1.00
T2	0.97	1.00	T2a	0.97	1.00	T3	0.97	1.00
T4a	0.97	1.00	T4b	0.99	1.00	T4c	0.99	1.00
T4d	0.98	1.00	T5a	1.00	1.00	T5b	1.00	1.00
T5c	0.99	1.00	T5d	0.99	1.00	Tm1	0.74	1.00
Tm2	0.93	1.00	Tm3	0.81	1.00	Tm4	0.99	1.00
Tm5Y	1.00	1.00	Tm5a	0.97	1.00	Tm5c	0.28	1.00
Tm16	0.99	1.00	Tm20	1.00	1.00	Tm28	0.75	0.99
Tm30	0.57	1.00	TmY3	1.00	1.00	TmY4	0.99	1.00
TmY5a	0.94	1.00	TmY9	0.95	1.00	TmY10	0.97	0.94
TmY13	0.99	1.00	TmY15	0.97	1.00	TmY18	0.99	1.00

Supplementary Table 5: **Single-type degenerate variants of the Flyvis weight matrix.** Each row reports $\mathbf{W}' = \mathbf{W} + \mathbf{\Delta}$, where $\mathbf{\Delta}$ is the maximum-amplitude ($8\times$ mean group weight) sum-zero perturbation within the degenerate groups of one cell type (52 types total). $R^2_{\mathbf{W}}$: weight-space agreement with ground truth; rollout Pearson r: voltage-trajectory agreement over 1,000 frames. Substantial weight deviation ($R^2_{\mathbf{W}} \in [0.28, 1.0]$) coexists with preserved dynamics ($r > 0.94$), confirming the structural null space of Eq. (12). TmY14 omitted (rollout diverged).

D Video stimulus in simulations

Visual inputs are derived from the DAVIS-2017 dataset [Pont-Tuset et al., 2017], a benchmark dataset for annotated natural videos. We used 151 videos (10,474 frames) in total, 80 videos (5,635 frames) of which were used for training (80%) and validation (20%), while 71 videos (4,839 frames) were withheld and solely used for reporting results in this manuscript. DAVIS sequences are augmented with horizontal/vertical flips and four rotations.

The frames are converted to grayscale, scaled to the range $[0, 1]$, and resampled onto the hexagonal photoreceptor lattice of 217 columns, giving 1,736 input neurons (8 photoreceptor types R1–R8 per column).

Null stimulus injection. For the simulation conditions reported in the main results, we prepend a zero-stimulus window to every sequence (see Supp. Fig. 4 for an example trace, in which the null window is highlighted in blue): the first 50% of each video clip is replaced by frames of constant zero luminance ($I_i(t) = 0$ for the corresponding photoreceptor inputs), while the remaining 50% contains the natural DAVIS frames. Concretely, for a sequence of length T we set $I_i(t) = 0$ for $t \in [0, T/2)$ and pass the original DAVIS frames during $t \in [T/2, T)$. The network's pre-stimulus warm-up at 0.5 luminance is left unchanged.

E Flyvis simulations with realistic FlyWire edges

Coregistration Lappalainen et al. [2024] compiled whole-eye connectivity from local reconstructions. The resulting Flyvis model connectome is a convolutional (translation-invariant) graph: Every

visual column shares the same set of $2,355$ edge templates, each parametrized by source type, target type, and column offset. This reductionist approach discards real synaptic connectivity variability across columns and anatomical location information of neurons and synaptic connections. To recover this information, we coregistered the Flyvis connectome to FlyWire with a pipeline including the following phases: We tiled the Flyvis convolutional template (65 cell types, 2,355 edge templates) across every reconstructed column of the right-optic-lobe in FlyWire. This resulted in $50{,}412$ candidate nodes and $1{,}650{,}671$ candidate edges for coregistration. Of these, 64.9% of nodes register to a FlyWire `root_id` and 48.5% of edges have both endpoints registered, jointly covering 93.5% of the $2{,}355$ templates and matching $3{,}800{,}254$ FlyWire synapses. The hybrid graph keeps FlyWire-observed edges wherever the postsynaptic neuron's anatomical receptive field was reconstructed in FlyWire and falls back to the Flyvis template edges elsewhere. It adds novel edges present only in FlyWire for type pairs that already exist in the Flyvis template. This results in two graphs: 'FlyWire eye' ($50{,}412$ neurons; $1{,}266{,}378$ edges) and "e8 hybrid", the 217-column restricted central portion of the eye ($13{,}741$ neurons; $327{,}358$ edges) (Supp. Tab. 3).

Because FlyWire synapse counts were systematically higher than Flyvis template values (median ratio $1.39\times$ at shared offsets), we applied a multiplicative per-type-pair mean-ratio correction to each FlyWire edge, so that edge weights remain compatible with the trained Flyvis scaling parameters and therefore ODE for simulation.

Connectivity uncertainty augmentation To model false-positive connectivity from electron-microscopy reconstruction errors, with anatomical location available from the coregistration, we constructed an augmented variant of the hybrid connectome in which every neuron pair within $\approx 6\,\mu$m ($\sim$one column in the optic lobe) inherits each other's edges with synapse count zero, across all type pairs. These null edges (n.e.) are spatially plausible but unmeasured connections that the GNN must learn to leave at zero or activate. The augmentation expands 'e8 hybrid' to $2{,}418{,}403$ edges and 'FlyWire eye' to $9{,}642{,}335$ edges and represents the worst case in which every confusable edge is included.

F GNN

F.1 Training and agentic HPO.

The optimised networks of the GNN are g_ϕ and f_θ of Eq. (2), both modelled as MLPs (ReLU activation, hidden dimension $= 80$, 3 layers, output size $= 1$). Other learnables are the two-dimensional latent vector $\mathbf{a}_i$ associated with each neuron and the connectivity matrix $\hat{\mathbf{W}}$. We implemented the GNNs in PyTorch Geometric [Fey and Lenssen, 2019] and used Adam gradient descent with separate parameter-group learning rates $\mathrm{lr}_W = 9{\cdot}10^{-4}$ (connectivity), $\mathrm{lr} = 1.8{\cdot}10^{-3}$ (g_ϕ, f_θ), $\mathrm{lr}_{\mathrm{emb}} = 2.3{\cdot}10^{-3}$ (embeddings), batch size 4, 1 nominal epoch with `data_augmentation_loop`$= 500$ over $64\,000$ training frames ($\approx 5{\cdot}10^5$ to $2{\cdot}10^6$ gradient steps across conditions). Time points are sampled uniformly at random at each step.

The agentic loop of Appx. G was run independently for each of the six conditions of Supp. Tab. 6 ($\sim$100 iterations of 4 parallel slots per condition, single-parameter mutations, 4-seed robustness validation of every candidate winner). All six converged on the same consistent conclusion about the role of each term of Eq. (4). The ℓ_1 prior on $\hat{\mathbf{W}}$ (γ_0) is the only important regularizer: setting $\gamma_0 = 0$ leaves rollouts nearly unchanged ($r = 0.999$) but collapses $R^2_{\hat{W}}$ from 0.97 to 0.39, a direct manifestation of the degeneracy discussed in Appx. C. γ_1, which controls ℓ_2 regularization, provides no measurable benefit. Among the g_ϕ priors, μ_1 prevents the trivial collapse $g_\phi \equiv 0$ and was selected in $[750, 2000]$ across conditions; μ_2 stabilises at 0.9 for the $\sigma = 0.05$ sub-conditions and is driven to 0 only in the noise-free condition; λ_1 takes a distinct value in the noise-free condition ($\lambda_1 = 0.14$, $+0.054$ on $R^2_{\hat{W}}$) versus a uniform 0.28 across the $\sigma = 0.05$ sub-conditions. The two f_θ priors split by noise regime: λ_0 (ℓ_1) is needed at the $\sigma = 0.05$ working point ($\lambda_0 = 0.05$, identical across the four $\sigma = 0.05$ sub-conditions) but is driven to 0 in the noise-free and $\sigma = 0.5$ extremes; λ_2 (ℓ_2) acts as the magnitude anchor of $\hat{\mathbf{W}}$ and stays at 10^{-3} for every noisy condition.

A clean cross-condition pattern emerged: the four σ=0.05 sub-conditions share an identical regulariser block, with the noise-free and σ=0.5 winners deviating at opposite poles. The σ=0.05 winner sits between the two extremes, is the closest single point to the per-coefficient medians of Supp.

Tab. 6, and is the only one validated under 12-seed robustness with zero catastrophic failures; we therefore retain it unchanged for every GNN result reported in the paper, including the FlyWire and degraded-connectome variants.

Recurrent training for the 1/5-frames row of Supp. Tab. 4. With only every fifth frame observed, consecutive targets are separated by $5\Delta t$ and the one-step regression used in every other condition no longer applies. We replace it with a $K = 5$-step truncated backpropagation-through-time (BPTT) rollout: the GNN is unrolled under its own predictions for five Euler steps from an observed frame, and gradients propagate through the full unroll to the next observed frame. To stabilise the unroll we raise μ_1 from 750 to 9,000 and lower μ_2 from 0.9 to 0.1, run batch size 1 and $n_{\text{epochs}} = 2$; all other hyperparameters of Supp. Tab. 6 are kept at the consensus values.

Supplementary Table 6: **GNN optimum hyperparameters** as tuned by the agentic loop (Appendix G) for each of the six evaluation conditions, plus the consensus configuration applied uniformly to every GNN result reported in the paper, including the FlyWire and degraded-connectome variants. The consensus is adopted from the low model noise ($\sigma = 0.05$) winner, the closest single point to the per-coefficient medians and the only one validated under 12-seed robustness with zero catastrophic failures. The agentic loop converged on a shared regulariser block ($\gamma_0 = 1.5 \times 10^{-4}$, $\mu_1 = 750$, $\mu_2 = 0.9$, $\lambda_1 = 0.28$, $\lambda_0 = 0.05$, $\lambda_2 = 10^{-3}$) across all four $\sigma = 0.05$ sub-conditions (clean, low measurement noise, false-positive edges, false-negative edges); only the per-condition learning rates and batch size differ. All coefficients refer to the loss of Eq. 4.

		learning rates			$\hat{\mathbf{W}}$ reg		g_ϕ reg			f_θ reg	
condition	bs	lr_W	lr	lr_{emb}	γ_0	γ_1	μ_1	μ_2	λ_1	λ_0	λ_2
noise-free	2	$3.0{\cdot}10^{-4}$	$9.0{\cdot}10^{-4}$	$1.6{\cdot}10^{-3}$	$1.5{\cdot}10^{-4}$	0	$1.5{\cdot}10^{3}$	0	0.14	0	0
low intr. noise	4	$9.0{\cdot}10^{-4}$	$1.8{\cdot}10^{-3}$	$2.3{\cdot}10^{-3}$	$1.5{\cdot}10^{-4}$	$1.5{\cdot}10^{-6}$	$7.5{\cdot}10^{2}$	0.9	0.28	0.05	$1.0{\cdot}10^{-3}$
high intr. noise	2	$1.5{\cdot}10^{-4}$	$1.2{\cdot}10^{-3}$	$1.6{\cdot}10^{-3}$	0	0	$7.5{\cdot}10^{2}$	0.9	0	0	$1.0{\cdot}10^{-3}$
low meas. noise	6	$9.0{\cdot}10^{-4}$	$1.8{\cdot}10^{-3}$	$2.3{\cdot}10^{-3}$	$1.5{\cdot}10^{-4}$	$1.5{\cdot}10^{-6}$	$2.0{\cdot}10^{3}$	0.9	0.28	0.05	$1.0{\cdot}10^{-3}$
+400% null edges	4	$9.0{\cdot}10^{-4}$	$1.8{\cdot}10^{-3}$	$2.5{\cdot}10^{-3}$	$1.5{\cdot}10^{-4}$	$1.5{\cdot}10^{-6}$	$7.5{\cdot}10^{2}$	0.9	0.28	0.05	$1.0{\cdot}10^{-3}$
−20% edges removed	4	$6.0{\cdot}10^{-4}$	$1.2{\cdot}10^{-3}$	$1.6{\cdot}10^{-3}$	$1.5{\cdot}10^{-4}$	$1.5{\cdot}10^{-6}$	$7.5{\cdot}10^{2}$	0.9	0.28	0.05	$1.0{\cdot}10^{-3}$
consensus (used)	**4**	$\mathbf{9.0{\cdot}10^{-4}}$	$\mathbf{1.8{\cdot}10^{-3}}$	$\mathbf{2.3{\cdot}10^{-3}}$	$\mathbf{1.5{\cdot}10^{-4}}$	$\mathbf{1.5{\cdot}10^{-6}}$	$\mathbf{7.5{\cdot}10^{2}}$	**0.9**	**0.28**	**0.05**	$\mathbf{1.0{\cdot}10^{-3}}$

F.2 Extraction of synaptic weights, time constants, and resting potentials

The GNN of Eq. (2) does not expose the membrane time constants $\hat{\tau}_i$, the resting potentials $\hat{V}_i^{\text{rest}}$, or a directly comparable per-edge synaptic weight as learnable parameters. The leaky-integrator term of Eq. (1) is absorbed into the postsynaptic update f_θ, while the per-edge weight $\hat{W}_{ij}$ that appears in the message is entangled with the gain of g_ϕ on the presynaptic side and the gain of f_θ on the postsynaptic side: a common rescaling of $\hat{\mathbf{W}}$ and the two MLPs leaves the predicted dynamics unchanged. We therefore extract the three biophysical quantities reported in the main paper in a post-processing step that linearises f_θ and g_ϕ around each neuron's working point on the same training trajectory $v_i(t)$ that drives the prediction loss. The extraction is run unchanged across every noise regime, FlyWire variant, and degraded-data condition.

Per-neuron working range. For every neuron i we summarise its trajectory by the empirical mean μ_i and standard deviation σ_i of $v_i(t)$ over the training horizon, and define the natural activity range

$$\mathcal{D}_i \;=\; \big[\mu_i - 2\sigma_i,\; \mu_i + 2\sigma_i\big]. \tag{13}$$

All affine fits of f_θ and g_ϕ are restricted to $\mathcal{D}_i$, so that the recovered slopes and offsets describe the learned dynamics where they are actually evaluated, rather than in extrapolation regions that the trained model never visits.

Time constant and resting potential. The leaky-integrator term of Eq. (1) is linear in v_i at fixed cell-type identity. We probe the postsynaptic update at zero aggregated input and zero stimulus,

$$\phi_i(v) \;:=\; f_\theta\big(v,\, \mathbf{a}_i,\, m = 0,\, I = 0\big), \tag{14}$$

sample ϕ_i on a dense grid of $v \in \mathcal{D}_i$, and fit a least-squares affine model

$$\phi_i(v) \;\approx\; s_i\, v + b_i. \tag{15}$$

Identifying Supp. Eq. 15 with the input-free, noise-free version of Eq. (1), $dv_i/dt = -v_i/\tau_i + V_i^{\text{rest}}/\tau_i$, gives the learned time constant and resting potential as

$$\hat{\tau}_i = -\frac{1}{s_i}, \qquad \hat{V}_i^{\text{rest}} = -\frac{b_i}{s_i}. \tag{16}$$

Neurons with a degenerate slope ($s_i \to 0$) and time constants that fall outside the physical range are clipped, and contribute to the outlier fraction reported alongside the parameter R^2.

Effective synaptic weights. The message in Eq. (2) couples each edge weight $\hat{W}_{ij}$ to two gain factors that the network is free to absorb: the rate of change of the rectified output of g_ϕ at the presynaptic working point and the rate of change of f_θ in its message channel at the postsynaptic working point. A common rescaling of $\hat{\mathbf{W}}$ and either gain leaves the predicted dynamics unchanged, so $\hat{\mathbf{W}}$ alone is not directly comparable to the simulator weight. The postsynaptic gain also folds in the learned membrane time constant: matching Eq. (2) term-by-term against the time-derivative form of Eq. (1) shows that the message channel of f_θ contributes $\hat{W}_{ij}\, g_\phi^2/\hat{\tau}_i$, so that the correction must include a factor of the learned $\hat{\tau}_i$ from Supp. Eq. 16.

We therefore linearise both functions on each neuron's activity range. For every presynaptic neuron j we evaluate the squared output $g_\phi(v, \mathbf{a}_j)^2$ that actually enters the message on the positive part of $\mathcal{D}_j$, fit it with an affine model, and record its slope η_j. For every postsynaptic neuron i we evaluate the partial derivative of f_θ with respect to its message argument at a small set of operating points sampled uniformly along the training trajectory and take the median across probes, yielding κ_i. The recovered effective weight on edge (i, j) is

$$\hat{\mathbf{W}}_{ij}^{\text{eff}} = \hat{\tau}_i\, \kappa_i\, \eta_j\, \hat{W}_{ij} = -\frac{\kappa_i\, \eta_j\, \hat{W}_{ij}}{s_i}, \tag{17}$$

where the second form uses $\hat{\tau}_i = -1/s_i$ from Supp. Eq. 16. The two gain factors and the learned time constant cancel the corresponding factors in Eq. (2) when g_ϕ^2 and the message channel of f_θ are well approximated by their linear fits on $\mathcal{D}_i$ and $\mathcal{D}_j$. The connectivity R^2 reported in the main paper and in Appx. I is computed between $\hat{\mathbf{W}}_{ij}^{\text{eff}}$ and the simulator weight W_{ij}, summed over every edge of the prior graph; for the degraded-connectivity variants of Supp. Tab. 4, this includes the augmented zero-weight edges, which the GNN optimization must drive to zero.

F.3 Joint stimulus recovery with SIREN.

The unkown-stimulus results of Supp. Tab. 4 is obtained by hiding the visual stimulus over the 1,736 retina neurons. To test whether the visual input $I_i(t)$ entering Supp. Eq. 18 can itself be recovered from activity alone, we extend the GNN with an implicit neural representation of the stimulus field $\hat{I}^*(t, x, y)$, modelled as a coordinate-based MLP with sine activations [SIREN, Sitzmann et al., 2020, input size $= 3$, hidden dimension $= 2048$, 4 hidden layers, output size $= 1$, $\omega_0 = 4096$, outermost-linear head]. The SIREN is queried at the 1 736 photoreceptor positions (x_i, y_i) at every time step and substitutes for $I_i(t)$ in the GNN forward pass. The GNN and the SIREN are jointly optimised on the same prediction loss but with strongly asymmetric learning rates and a two-phase "alternate-training" schedule: epoch 0 uses the full GNN learning rates above with $\text{lr}_{\text{SIREN}} = 10^{-8}$; epochs 1–2 scale the GNN learning rates by 0.05 while lr_{SIREN} is held constant, letting the SIREN catch up on a stable graph backbone.

G Agent-in-the-loop parameter optimisation

The optimisation space combining simulation parameters (noise σ, connectivity constraints, n_{frames}) and machine learning parameters (e.g. for GNN, $\lambda_0, \lambda_1, \lambda_2, \lambda_3, \gamma_0, \gamma_1, \mu_0, \mu_1, \mu_2$, learning rates, etc.) is too large for grid search. We instead deploy a closed loop coupling three components: a GNN training engine, a large language model agent (Claude Code, Anthropic), and an explicit structured summary. The system operates through text file exchange. The instruction file defines the inverse problem, the quantitative score objective, the metrics to track, the parameter ranges, and the logging format. The structured summary maintains accumulated understanding in two tiers: established principles and open questions that persist across blocks, and a working record of current hypotheses

and recent iterations. A configuration file is edited by the agent to set the next experiment; a results file is written by the experiment engine and read by the agent.

The primary objective of the agentic loop is to optimise hyperparameters against a quantitative score ($R^2_{\mathbf{W}}$ or rollout accuracy). But we also constrain the agent to hypothesis-driven protocol: a testable hypothesis must be stated before each experiment, and only experimental outcomes can validate or falsify it. The next experiment is therefore selected by the hypothesis the agent currently seeks to test rather than by a numerical exploration/exploitation trade-off.

Experiments are organised into blocks of 8-16 iterations (multiples of 4 to match the parallel batch size). The agent does not start from scratch, it is initialised with a configuration established from prior works. The simulation regime of each block can be specified by the user in the instruction file or chosen by the agent. Within a block, the agent explores GNN training parameters guided by its current hypothesis. To make the protocol robust, we run 4 experiments in parallel and submit the resulting batch of training outcomes to the agent. Reasoning over a batch rather than a single trial sharpens both confirmation and falsification. The structured summary also serves as an explicit long-term memory, overcoming the finite context window of the agent and enabling accumulation across hundreds of iterations. With this protocol, convergence to the optimal configuration is reached within ~ 100 GNN training experiments sweeping sometimes more than 10 parameters.

We initially adopted UCB tree search as in AlphaEvolve and FunSearch [Romera-Paredes et al., 2024, Novikov et al., 2025], where parameter mutations are sampled from a tree balancing exploitation and exploration. Once the hypothesis-driven constraint was introduced, we found that the agent stopped consulting the UCB scores: parent selection and parameter mutations were driven entirely by hypotheses formulated from the structured summary and recent experimental results. We accordingly removed the UCB component, which had become redundant.

As an example, the instruction file passed to the agent for the noise-free condition is

```
# Flyvis GNN - Connectome Recovery (noise-free)

## Goal

Optimize GNN hyperparameters for maximum **connectivity matrix recovery (conn_R2)** on Flyvis with
no noise (sigma=0.0). Two sub-goals:

- Maximize conn_R2 (primary): recover synaptic weight matrix W from clean neural activity
- Understand the noise-free ceiling: why is conn_R2=0.923 (prior winner) lower than noise_005's 0.982?

The exploration starts from the **default config** below to enable before/after comparison with
the prior winner (conn_R2=0.923, CV=0.82%).

## Scientific Context

**Core research question: Why does noise-free training produce lower connectome recovery (0.923)
than sigma=0.05 training (0.982)? Does noise act as regularization?**

The noise-free case is surprising - removing noise should make W recovery easier, yet the prior winner
is lower. Key hypotheses to test:
- Noise acts as implicit regularization (Langevin dynamics): sigma=0.05 prevents over-fitting to W
- The embedding_dim=4 (unique to noise-free) may be under-exploited - extra dims could help or hurt
- All regularization is 0 in the default config - should some regularization be introduced?
- coeff_g_phi_diff=1500 is 2x higher than noisy variants - explore if this is optimal
- The absence of noise sharpens the optimization landscape, potentially creating narrow basins

Key differences from noisy variants:
- embedding_dim=4 (vs 2 for noise_005 and noise_05)
- All L1/L2 regularization = 0 (vs non-zero for noisy variants)
- coeff_g_phi_norm = 0 (vs 0.9 for noisy variants)
- coeff_g_phi_diff = 1500 (vs 750 for noisy variants)

## Noise Model

v_i(t+1) = v_i(t) + dt * f(v_i(t), W, a_i, I_i(t))
sigma = 0.0 (noise_model_level)

No noise is added to training data. Train and test conditions are identical.

## Metrics

During test/validation:
- **PRIMARY METRIC: `conn_R2`** (higher is better; R^2 of learned W vs ground-truth W)
- `tau_R2`: R^2 of tau (time constant) recovery
- `V_rest_R2`: R^2 of V_rest (resting potential) recovery
- `cluster_accuracy`: cell-type clustering accuracy from neuron embeddings
- `rollout_pearson_r`: Pearson r of autoregressive rollout vs ground truth

**Robustness classification** (4 seeds per iteration):
- **Stable-Robust**: all 4 seeds conn_R2 >= 0.90, CV < 3%
- **Stable**: mean conn_R2 >= 0.85, CV < 10%
```

```
- **Unstable**: mean < 0.85 OR CV >= 10%
- **Catastrophic**: any seed conn_R2 < 0.50

## Scientific Method

Strict **hypothesize -> test -> validate/falsify** cycle:

1. **Hypothesize**: Form a specific, testable prediction
2. **Design experiment**: Change **EXACTLY ONE** parameter at a time (causality rule)
3. **Run training**: 4 slots (1 control + 3 experiments in EXPLORATION; 4 same config in ROBUSTNESS)
4. **Analyze results**: Use conn_R2 AND cluster_accuracy to understand embedding quality
5. **Update understanding**: Revise hypotheses based on evidence

**CRITICAL**: You can only hypothesize. Only training results validate or falsify.

### CAUSALITY RULE (MANDATORY)

**If you change more than one parameter per slot, you CANNOT attribute the effect.**

- In EXPLORATION mode: Slot 0 = parent/baseline (unchanged control). Slots 1-3 each change
  **exactly one** parameter from the parent.
- In ROBUSTNESS mode: all 4 slots use the same config (different seeds test robustness).

## Flyvis Model

tau_i * dv_i/dt = -v_i + V_rest_i + sum_j W_ij * g(v_j) + I_i(t)

- **13,741 neurons**, 65 cell types, **434,112 edges**
- **1,736 input neurons** (photoreceptors, DAVIS visual input)
- Noise level: sigma=0.0 (noise-free)
- 64,000 frames, delta_t = 0.02

## GNN Architecture

g_phi(v_j, embed_j) -> message_ij          (edge MLP, per-edge messages)
sum_j W_ij * g_phi(v_j) -> agg_i           (weighted aggregation)
f_theta(v_i, agg_i, embed_i) -> dv_i/dt    (node update MLP)

- Per-neuron embedding: learnable `embedding_dim`-dim vector
- **embedding_dim=4** (unique to noise-free; 2x larger than noisy variants)
- `g_phi_positive=true`: g_phi output clipped to [0, inf)

## GNN Architecture Parameters

| Parameter       | Default | Description                                          |
| --------------- | ------- | ---------------------------------------------------- |
| hidden_dim      | 80      | Width of hidden layers in g_phi and f_theta          |
| n_layers        | 3       | Depth of g_phi and f_theta networks                  |
| embedding_dim   | 4       | Per-neuron learnable embedding dimension             |

## Training Parameters

| Parameter                | Default      | Description                                          |
| ------------------------ | ------------ | ---------------------------------------------------- |
| lr_W                     | 0.0006       | Learning rate for W matrix                           |
| lr                       | 0.0018       | Learning rate for g_phi and f_theta MLP weights      |
| lr_embedding             | 0.00155      | Learning rate for per-neuron embeddings              |
| data_augmentation_loop   | 30           | Augmentation loops per epoch (DAL)                   |
| batch_size               | 4            | Samples per batch                                    |
| coeff_g_phi_diff         | 1500         | L2 penalty against g_phi collapse (2x noisy)         |
| coeff_g_phi_norm         | 0.0          | L2 norm reg. on g_phi (0 for noise-free)             |
| coeff_g_phi_weight_L1    | 0.0          | L1 weight reg. on g_phi (0 for noise-free)           |
| coeff_f_theta_weight_L1  | 0.0          | L1 weight reg. on f_theta                            |
| coeff_f_theta_weight_L2  | 0.0          | L2 weight reg. on f_theta                            |
| coeff_W_L1               | 0.0          | L1 reg. on W                                         |
| coeff_W_L2               | 0.0          | L2 reg. on W                                         |
| regul_annealing_rate     | 0.0          | MUST be 0.0 with n_epochs=1                          |
| w_init_mode              | randn_scaled | W init: randn_scaled / zeros / uniform_scaled        |
| w_init_scale             | 1.0          | Scale for randn init (bound = scale/sqrt(n_edges))   |

**Hard runtime limit (120 min)**: Cluster enforces 120-min wall-clock limit.

**Fixed: n_epochs=1** - do not change.

## Block Structure

| Block | Focus                  | Parameters to scan                                                    |
| ----- | ---------------------- | --------------------------------------------------------------------- |
| 1     | Baseline robustness    | All 4 slots = default config (ROBUSTNESS)                             |
| 2     | Learning rates         | lr_W, lr, lr_embedding                                                |
| 3     | g_phi diff + norm      | coeff_g_phi_diff, coeff_g_phi_norm                                    |
| 4     | Regularization intro   | coeff_g_phi_weight_L1, coeff_f_theta_weight_L1, coeff_W_L1            |
| 5     | Architecture           | embedding_dim, hidden_dim                                             |
| 6     | Training regime        | batch_size, data_augmentation_loop                                    |
| 7     | Free exploration       | Combine best from Blocks 2-6                                          |
| 8     | Final robustness       | Best config, 4 slots same (ROBUSTNESS)                                |
| 9     | CV robustness          | Best config, 8 seeds over 2 iterations (generate_data=true)           |

## Iteration Workflow

### Step 1: Read Working Memory + User Input
### Step 2: Analyze Results (4 slots)
```

```
For each slot: read conn_R2, tau_R2, V_rest_R2, cluster_accuracy, rollout_pearson_r;
compare to the noise-free baseline (0.923) and to noise_005 (0.982); classify.

### Step 3: Write Log Entry + Update Memory

## Iter N: [stable_robust/stable/unstable/catastrophic]
Node: id=N, parent=P
Hypothesis tested: "[quoted hypothesis]"
Config: lr_W=X, lr=Y, lr_emb=Z, DAL=D, bs=B,
        g_diff=A, g_norm=B, g_L1=C, f_L1=D, W_L1=E,
        w_init=F, emb_dim=G, hidden=H
Slot 0..3: conn_R2, tau_R2, Vr_R2, cluster_acc, rollout_r, sim_seed, train_seed
Seed stats: mean_conn_R2, std, CV%, catastrophic=N/4
Mutation: [param]: [old] -> [new]
Verdict: [supported/falsified/inconclusive]
Next: parent=P

### Step 4: Acknowledge User Input
### Step 5: Formulate Next Hypothesis + Edit 4 Config Files

## Start Call

When prompt says PARALLEL START:
- Slot 0 = default config (before-exploration baseline):
  lr_W=0.0006, lr=0.0018, lr_embedding=0.00155, batch_size=4, DAL=30
  coeff_g_phi_diff=1500, coeff_g_phi_norm=0.0, coeff_g_phi_weight_L1=0.0
  coeff_f_theta_weight_L1=0.0, coeff_W_L1=0.0, w_init_mode=randn_scaled, embedding_dim=4
- Block 1 is ROBUSTNESS mode: Slots 1-3 also use the same default config (different seeds)
- Hypothesis: "The default noise-free GNN config achieves conn_R2 >= 0.90 with CV < 5% but
  remains below noise_005's winner (0.982). Prior winner: conn_R2=0.923, CV=0.82%."
```

As an example, the structured summary built by the agent for the noise-free condition is

```
# Working Memory: Flyvis_noise_free

## Paper Summary (update at every block boundary)

**GNN optimization** (EXPLORATION COMPLETE -- 108/108 iters, Blocks 1-9):
Champion config achieves conn_R2=0.9665 single-seed ATB (Iter 77), 24-seed combined
mean=0.9453+/-0.0159 (CV=1.68%, Block 8+9), STABLE-ROBUST, beating the prior winner
(0.923) on 22/24 seeds. Key hyperparameters: g_phi_wL1=0.14 (single largest lever:
+0.054 over baseline, narrow peak), W_L1=1.5e-4, lr_W=3e-4, lr=9e-4, DAL=150,
n_epochs=1, bs=4, g_diff=1500, g_norm=0, randn_scaled, emb_dim=2, hidden=80.

**LLM-driven exploration** (EXPLORATION COMPLETE):
All 21 falsified hypotheses and 21 established principles are closed; champion
config is definitively optimal. Gap vs noise_005 narrowed from 0.085 to 0.0155
(ATB 0.9665 vs 0.982); structural noise-free ceiling at ~0.945 mean confirmed by
both Block 8 (fixed data, CV=1.62%) and Block 9 (generate_data=true, CV=1.74%).
Noise acts as implicit regularizer: noise-free requires explicit g_phi_wL1=0.14
to compensate, contrasting with the broad optimization basin that noise provides.

---

## Knowledge Base

### Robustness Comparison Table

| Iter        | Config summary                       | conn_R2          | CV%   | Verdict          |
| ----------- | ------------------------------------ | ---------------- | ----- | ---------------- |
| 1-12        | 12 seeds Block 1                     | 0.913+/-0.042    | 4.6%  | Stable           |
| 13-16       | lr_W scan                            | 0.924-0.948      | -     | lr_W=3e-4 best   |
| 17-20       | lr scan                              | 0.900-0.953      | -     | lr=9e-4 best     |
| 21-24       | lr_emb scan                          | 0.868-0.955      | -     | Inconclusive     |
| 25-28       | g_diff scan                          | 0.922-0.951      | -     | g_diff=1500 best |
| 29-32       | g_norm scan                          | 0.905-0.955      | -     | g_norm=0 best    |
| 33-36       | g_norm=0, 4 seeds robustness         | 0.9446+/-0.012   | 1.3%  | STABLE-ROBUST    |
| 37-40       | W_L1 + g_phi_wL1 init                | 0.916-0.950      | -     | W_L1=1.5e-4 best |
| 41-44       | W_L1 extend + f_theta_wL1            | 0.860-0.939      | -     | f_theta_wL1=0.05 |
| 45-48       | W_L1 finalize, g_phi_wL1=0.28        | 0.869-0.967      | -     | W_L1=1.5e-4 peak |
| 49-60       | Architecture (emb_dim, hidden)       | 0.744-0.953      | -     | emb_dim=4 (NB: deployed=2) |
| 71          | g_phi_wL1=0.14 half-dose             | 0.9579           | -     | ATB raw_W=0.8106 |
| 73          | Block 7 CTRL g_phi_wL1=0.14          | 0.9630           | -     | ATB raw_W=0.8340 |
| 77          | Block 7 CTRL g_phi_wL1=0.14 (3rd)    | 0.9665           | -     | NEW ATB          |
| 78          | DAL=175                              | 0.9516           | -     | FALSIFIED        |
| 79          | g_phi_wL1=0.21                       | 0.9368           | -     | FALSIFIED        |
| 82          | W_L2=1e-5                            | 0.9362           | -     | FALSIFIED        |
| 84          | g_phi_norm=0.01                      | 0.9439           | -     | NEUTRAL/decline  |
| 85-96       | Block 8 (12-seed mean, fixed data)   | 0.9442+/-0.0153  | 1.62% | STABLE-ROBUST    |
| 97-108      | Block 9 (12-seed mean, gen_data=t)   | 0.9465+/-0.0165  | 1.74% | STABLE-ROBUST    |

ATB (single-seed, clean): Iter 77 conn_R2=0.9665.
Combined 24-seed: mean=0.9453, CV=1.68%, 0/24 catastrophic. EXPLORATION COMPLETE.

### Established Principles

1. DAL=150 outperforms DAL=30: +0.031 mean conn_R2, CV halved.
2. tau_R2 and Vr_R2 unreliable for Flyvis_A: model absorbs them implicitly.
3. Conservative LRs win noise-free: lr_W=3e-4 (0.5x default) +0.025;
   lr=9e-4 (0.5x default) +0.017. Hard cliff above lr=3.6e-3.
```

```
4. g_phi defaults (g_diff=1500, g_norm=0) both optimal for noise-free.
   g_norm=0 (vs 0.9 optimal for noise_005). g_diff=1500 (2x noisy variants).
5. W_L1=1.5e-4 is optimal W regularization. Block 4 scan: 0->5e-5->1.5e-4
   ->3e-4->5e-4. Peak at 1.5e-4 (+0.035 vs 0). Over-reg above 1.5e-4.
6. f_theta_wL1 beneficial range is narrow (~0.05). Dose: 0->0.05(+0.033)
   ->0.2(-0.082).
7. g_phi_wL1=0.14 is the OPTIMAL dose. Full dose-response:
   0.07(0.9292) < 0.14(ATB=0.9665) > 0.21(0.9368) > 0.28(~0.93-0.95)
   > 0.5(0.9168). Non-monotone sharp peak at 0.14.
8. n_epochs=2 same budget is HARMFUL for noise-free: raw_W_R2=0.370 vs 0.613.
9. g_phi_wL1 + f_theta_wL1 combination is ALWAYS ANTAGONISTIC: 0.28+0.05
   catastrophic (Iter 70); 0.14+0.05 -> -0.051 (Iter 74).
10. DAL=175 is over-training for noise-free: -0.015 vs DAL=150.
11. f_theta_msg_diff=50 is NEUTRAL for noise-free W recovery: -0.007.
12. W_L2=1e-5 is mildly harmful for noise-free: -0.014.
13. batch_size=2 is NEUTRAL for conn_R2: 2-seed mean=0.9577 ~ ctrl=0.9599.
14. g_phi_norm is harmful at ALL tested values (0.01-1.5): g_norm=0 optimal.
15. dale_law_score predicts conn_R2 (Block 8 confirmed, decouples with
    generate_data=true).

### Falsified Hypotheses

1. "g_phi_norm helps noise-free" -- ALL values 0.01-1.5. g_norm=0 optimal.
2. "W_L1=5e-4 beats W_L1=1.5e-4" -- FALSIFIED. Peak at 1.5e-4.
3. "f_theta_wL1=0.2 improves over 0.05" -- FALSIFIED. Over-reg.
4. "Larger emb_dim monotonically improves clustering" -- FALSIFIED.
   emb_dim=8 harms cluster_acc -0.073.
5. "emb_dim=6 improves conn_R2 over emb_dim=4" -- DEFINITIVELY FALSIFIED.
6. "n_epochs=2 same budget matches n_epochs=1" -- FALSIFIED. raw_W_R2
   catastrophically degraded.
7. "g_phi_wL1=0.28 + f_theta_wL1=0.05 combination is additive" --
   CATASTROPHICALLY FALSIFIED (conn_R2=0.0000, loss=47K).
8. "g_phi_wL1=0.07 is superior or equal to 0.14" -- FALSIFIED.
9. "DAL=175 fills budget and improves W convergence" -- FALSIFIED.
10. "g_phi_wL1=0.21 is between 0.14 and 0.28 performance" -- FALSIFIED.
    Peak is sharp and narrow.
11. "W_L2=1e-5 complements W_L1=1.5e-4 for noise-free" -- FALSIFIED.
12. "batch_size=2 improves conn_R2 over batch_size=4" -- NEUTRAL/FALSIFIED.

### Open Questions

1. Why does noise-free underperform vs noise_005? Gap narrowed to 0.0155.
   Remaining gap confirmed as structural limit: sharp narrow peak at
   g_phi_wL1=0.14. No remaining untested levers.

---

## Previous Block Summaries

### Block 5 Summary -- Architecture Search (iters 49-60)
emb_dim=4 OPTIMAL among tested values; emb_dim=6 FALSIFIED -0.005;
emb_dim=8 HARMS cluster -0.073. hidden=128 CONFIRMED +0.050 cluster_acc,
neutral conn_R2. Architecture NOT the bottleneck.

### Block 6 Summary -- Regularization Refinement (iters 61-72)
g_phi_wL1 dose-response COMPLETE: peak at 0.14 (ATB=0.9579, raw_W_R2=0.8106
HIGHEST EVER). 0.28 over-regularizes. g_phi_wL1=0.28+f_theta_wL1=0.05
COMBINATION CATASTROPHIC (conn_R2=0). f_theta_wL1=0.05 confirmed
individually (+0.024). n_epochs=2 same budget FALSIFIED.

### Block 7 Summary -- Free Exploration (iters 73-84)
ATB: Iter 77 conn_R2=0.9665. 4-seed champion (Iters 71,73,77,81):
mean=0.9594, CV=0.76%. All 9 tested levers FALSIFIED or NEUTRAL: f_theta_wL1
combo (ANTAGONISTIC), g_phi_wL1 neighbors (SHARP PEAK), DAL=175 (OVERFIT),
f_theta_msg_diff=50 (NEUTRAL), W_L2 (FALSIFIED), batch_size=2 (NEUTRAL),
g_phi_norm=0.01 (NEUTRAL).

### Block 8 Summary -- Final Robustness (iters 85-96)
STABLE-ROBUST CONFIRMED on 12 independent seeds. Mean=0.9442+/-0.0153,
CV=1.62%, 0/12 catastrophic. 10/12 seeds beat prior winner (0.923).
Key finding: dale_law_score is strong predictor of final conn_R2.

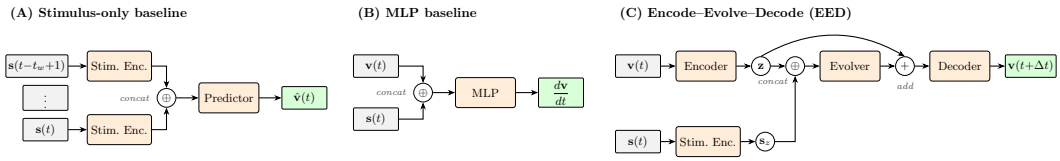
### Block 9 Summary -- CV Robustness (iters 97-108) -- COMPLETE
STABLE-ROBUST with generate_data=true on 12 seeds. Mean=0.9465+/-0.0165,
CV=1.74%, 0/12 catastrophic, 1/12 below prior winner. Performance matches
Block 8: generate_data=true does NOT degrade performance.

---

## EXPLORATION COMPLETE -- 108/108 Iterations, 9 Blocks

**Final Champion Config**:
- lr_W=3e-4, lr=9e-4, lr_emb=1.55e-3, DAL=150, n_epochs=1, bs=4
- g_diff=1500, g_norm=0, g_phi_wL1=0.14, f_L1=0, W_L1=1.5e-4, W_L2=0
- randn_scaled, emb_dim=2, hidden=80

**Final Statistics**:
- ATB (single-seed): Iter 77 = 0.9665
- Block 8 (12-seed, fixed data): mean=0.9442, CV=1.62%
- Block 9 (12-seed, generate_data=true): mean=0.9465, CV=1.74%
- Combined 24-seed: mean=0.9453, CV=1.68%, 22/24 beat prior winner (0.923),
  0/24 catastrophic
```



Supplementary Figure 13: Architectures of the three black box neuronal activity prediction baseline models discussed. Grey boxes are model inputs, green boxes are model outputs and orange boxes are sub-networks that are variations on the basic MLP architecture. **(A) Stimulus-only baseline** A length-t_w window of past stimuli is encoded frame-by-frame by a *shared* stimulus encoder; the concatenated latents are fed to a predictor MLP that outputs $\hat{\mathbf{v}}(t)$ with no dependence on past voltage. Since the Flyvis model was optimized for optical flow we expect that the neural activity at any time contains a representation of past stimuli. $t_w = 1$ serves as a baseline to captures the extent to which the stimulus alone can predict the instantaneous voltage. **(B) MLP baseline.** Voltages $\mathbf{v}(t) \in \mathbb{R}^N$ and stimulus $\mathbf{s}(t) \in \mathbb{R}^{N_{\text{in}}}$ are concatenated and mapped to $d\mathbf{v}/dt$ by a single MLP (L hidden layers, activation ϕ, ResNet-style skip connections). No graph structure is used. **(C) Encode–Evolve–Decode (EED).** A voltage encoder maps $\mathbf{v}(t)$ to a latent $\mathbf{z}(t)$; a stimulus encoder maps $\mathbf{s}(t)$ to $\mathbf{s}_z(t)$. A residual evolver updates the latent as $\mathbf{z}' = \mathbf{z} + \text{Evolver}([\mathbf{z}; \mathbf{s}_z])$ (evolver zero-initialised, so the map starts as identity), and a decoder produces $\mathbf{v}(t+\Delta t)$. All four sub-networks are MLPs with per-layer input-skip concatenations (NeRF-/DeepSDF-style). No graph structure is used.

```
- Gap vs noise_005 (ATB): 0.0155 -- structural noise-free ceiling confirmed

**Key finding**: g_phi_weight_L1=0.14 is the single most impactful lever
(+0.054 over baseline, sharp narrow peak). Noise acts as implicit regularizer
-- noise-free requires explicit regularization to compensate.
```

H Benchmark models

We describe below four baseline models that are distinctive in their architecture and inductive bias. The Known-ODE baseline incorporates maximal knowledge about the underlying system and sets an upper bound on the GNN, which assumes no knowledge of the underlying non-linearity and only knowledge of the non-zero edges in the connectome graph. The remaining models, depicted in Supp. Fig. 13, make different assumptions about the dynamics and stimuli. All of them assume that the stimuli are intrinsically low-dimensional, which is a reasonable assumption for natural images drawn from the DAVIS dataset. Stimulus-driven MLP is the most general: only making the assumption that neural activity contains some encoding of past stimuli. Recurrent MLP assumes that the neural activity can be described as a first order dynamical system whose variables are the activities themselves. EED model is a state space model that enforces that the neural dynamics is intrinsically low-dimensional, an active area of research starting from the early work of [Churchland et al., 2012, Pandarinath et al., 2018]. We show that despite the Recurrent MLP and EED model models being first order models of neural activity like the GNN (and Known-ODE) and accurately reproducing the auto-regressive rollout on held out test data, they do not capture the *true* dynamics. The binary connectome edges provide a very specific prior that allows the GNN to infer the true dynamics and models like the above lack.

H.1 Known-ODE

The known-ODE model fixes the functional form of the dynamics to the ground-truth Flyvis equation and learns only the biophysical parameters and the connectivity:

$$\hat{\tau}_i \frac{d\hat{v}_i(t)}{dt} = -\hat{v}_i(t) + \hat{V}_i^{\text{rest}} + \sum_{j \in \mathcal{N}_i} \hat{W}_{ij} \,\text{ReLU}\big(v_j(t)\big) + I_i(t). \tag{18}$$

The learnable parameters are the per-edge weights $\hat{W}_{ij}$, the per-neuron time constants $\hat{\tau}_i = \text{softplus}(\tilde{\tau}_i)$ (with $\tilde{\tau}_i \in \mathbb{R}$ the underlying optimized parameter), and the per-neuron resting potentials $\hat{V}_i^{\text{rest}}$. During training, inputs $I_i(t)$, adjacency $\mathcal{N}_i$, and activity $v_i(t)$ are given; $\hat{\mathbf{W}}, \tilde{\boldsymbol{\tau}}, \hat{\mathbf{V}}^{\text{rest}}$ are optimized by minimizing the difference between simulator $y_i(t) = dv_i(t)/dt$ and known-ODE

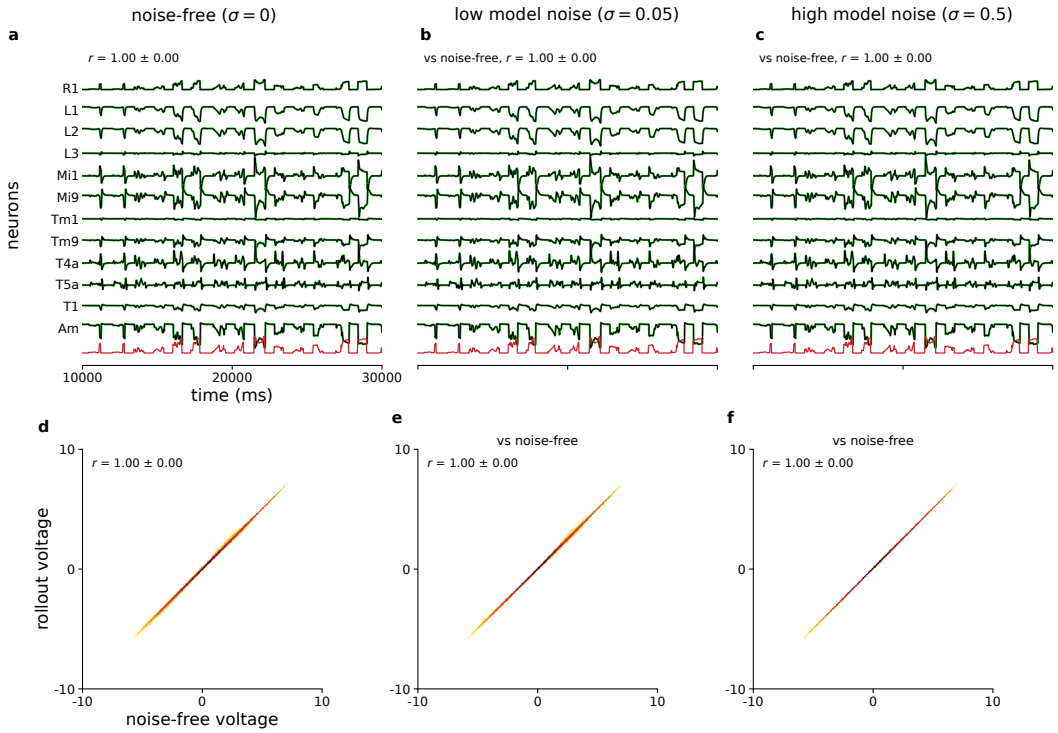
predictions $\hat{y}_i(t) = d\hat{v}_i(t)/dt$,

$$\mathcal{L}_{\text{pred}} = \sum_{i,t} \|\hat{y}_i(t) - y_i(t)\|_2. \tag{19}$$

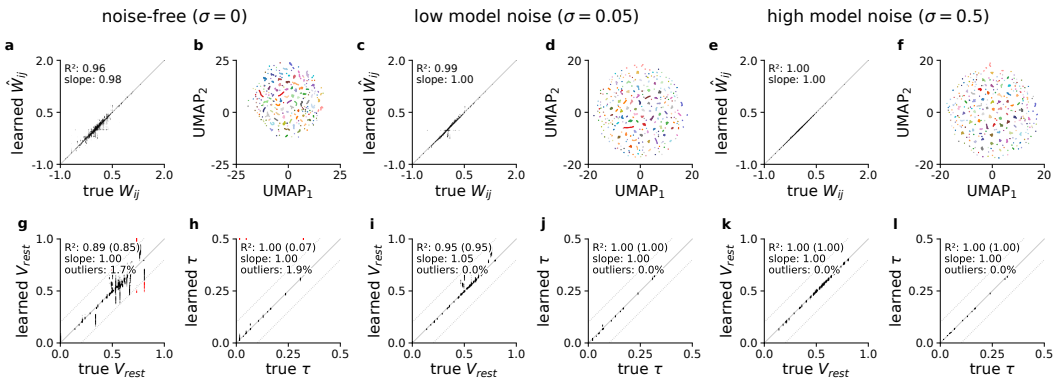
Even with the ODE structure fixed, recovery of $\hat{\mathbf{W}}$ from voltage traces remains non-unique (Appx. C). To constrain the solution space, we added regularization terms to $\mathcal{L}_{\text{pred}}$:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \gamma_0 \|\hat{\mathbf{W}}\|_1 + \eta_1 \|\tilde{\boldsymbol{\tau}}\|_2 + \eta_3 \|\hat{\mathbf{V}}^{\text{rest}}\|_2. \tag{20}$$

Note that the regularization we apply here differs from the regularization applied in Appx. C; there we find the minimum norm solution whose components are zero along the degenerate directions. The ℓ_1 term γ_0 on $\hat{\mathbf{W}}$ promotes sparse effective connectivity, the ℓ_2 penalty η_1 on $\tilde{\boldsymbol{\tau}}$ prevents drift of the time constants, and the ℓ_2 penalty η_3 on $\hat{\mathbf{V}}^{\text{rest}}$ penalizes large biases. The coefficients were tuned via the agentic optimization procedure described in Appx. G, which set the ℓ_2 term on $\hat{\mathbf{W}}$ and the ℓ_1 terms on $\tilde{\boldsymbol{\tau}}$ and $\hat{\mathbf{V}}^{\text{rest}}$ to zero; they are left out of Supp. Eq. 20. In practice we retain a single regularization profile for every Known-ODE result reported in the paper, namely the best paremeters found by the agentic exploration run on noise-free data (Appx. G): $\gamma_0 = 10^{-5}$ (ℓ_1 on $\hat{\mathbf{W}}$), $\eta_1 = 3\times10^{-3}$ (ℓ_2 on $\tilde{\boldsymbol{\tau}}$), $\eta_3 = 10^{-3}$ (ℓ_2 on $\hat{\mathbf{V}}^{\text{rest}}$) [the remaining three coefficients, zero, are left out of Supp. Eq. 20]. The same coefficients are applied unchanged to the noisy Flyvis conditions and to the FlyWireRF runs.



Supplementary Figure 14: **Known-ODE rollout prediction.** Known-ODE models (the published checkpoints, fold cv00, re-analysed with the current analysis code) evaluated on central 217-column Flyvis data under three model noise regimes ($\sigma = 0$, $\sigma = 0.05$, $\sigma = 0.5$), tested on held-out stimuli, and compared against the noise-free Flyvis simulation of the same stimuli. Top row (**a**, **b**, **c**): Known-ODE rollout traces for 12 representative cell types over a 20 s window (1,000 frames at $\Delta t = 20$ ms). Green: noise-free ground-truth voltage; black: Known-ODE rollout prediction; red: the visual input of the photoreceptor row. Bottom row (**d**, **e**, **f**): rollout voltage against noise-free ground-truth voltage, pooled over all (neuron, frame) pairs of the 7,207-frame rollout as a log-density image. Pearson r: per-neuron, Fisher-z pooled, on the full rollout.



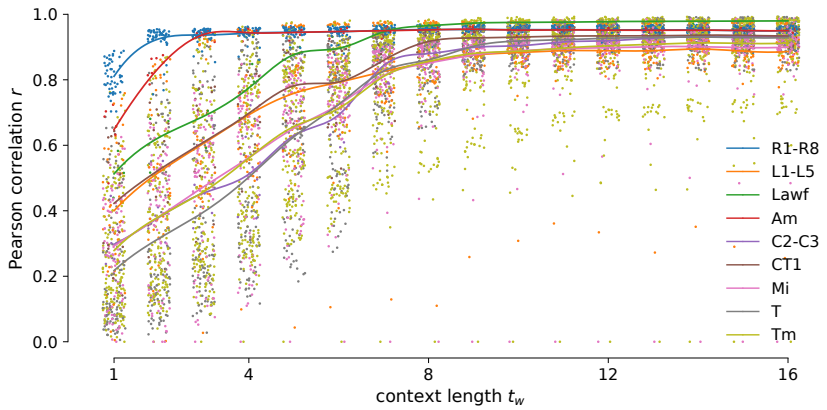
Supplementary Figure 15: **Known-ODE parameter recovery**. Known-ODE models (the published runs, fold cv00, re-analysed with the current analysis code) on Flyvis-217 under three model noise regimes ($\sigma = 0, 0.05, 0.5$). **(a, c, e)** Learned against true synaptic weight $\hat{W}_{ij}$, one point per edge, all 434,112 edges (the Known-ODE learns W directly); 1,085 edges (0.25%) lie outside the $[-1, 2]$ axes. **(b, d, f)** UMAP projection of the per-neuron feature vector the clustering accuracy is computed on (learned τ and V^{rest}, with the mean, SD, minimum and maximum of the incoming and of the outgoing learned weights, standardised), coloured by ground-truth cell type. **(g, i, k)** Resting potentials V_i^{rest}. **(h, j, l)** Time constants τ_i. Each scatter is annotated with the R^2 and slope of learned against true quoted in Tab. 1 (identity-line R^2; for τ and V^{rest} the inlier R^2 with the all-neuron R^2 in parentheses, outliers beyond the dotted ± 0.1 s / ± 0.2 bands in red, drawn on the axis edge when beyond it, and their fraction printed).

Supplementary Table 7: **Known-ODE evaluation across degraded versions** of the Flyvis training data: model and measurement noise, added/removed connectivity edges. The published Known-ODE runs, re-analysed with the paper's current analysis code. Five-fold cross-validation (mean ± SD). Prediction: metrics computed on noise-free data with held-out stimuli (7,207–7,999 frames by fold). Parameter recovery: $R^2_{\hat{W}}$ by the template readout over the fitted edges, as for the GNN rows; $R^2_{\hat{\tau}}$ and $R^2_{\hat{V}^{\text{rest}}}$ over all neurons, outlier-corrected (outlier fraction in parentheses, Appendix I); GMM clustering accuracy over 65 cell types. Green: value > 0.9. Orange: value < 0.3.

condition	model noise σ	measurement noise γ	edges	prediction one-step r	prediction rollout r	parameter recovery $R^2_{\hat{W}}$	parameter recovery $R^2_{\hat{\tau}}$ (out.%)	parameter recovery $R^2_{\hat{V}^{\text{rest}}}$ (out.%)	cluster acc.
noise-free	0	0	434 112	1.00±0	1.00±0	0.97±6E−4	1.00±3E−5 (2)	0.90±2E−3 (2)	0.92±2E−3
low model noise	0.05	0	434 112	1.00±0	1.00±0	0.99±2E−4	1.00±8E−6 (0)	0.95±1E−3 (0)	0.93±5E−3
high model noise	0.5	0	434 112	1.00±0	1.00±0	1.00±2E−6	1.00±7E−7 (0)	1.00±9E−6 (0)	0.93±5E−3
low meas. noise	0.05	0.1	434 112	0.99±0	0.96±2E−3	0.73±8E−4	0.31±4E−3 (8)	0.70±1E−3 (29)	0.88±0.01
mid meas. noise	0.05	0.2	434 112	0.96±9E−4	0.92±3E−3	0.47±1E−3	0.06±0.01 (10)	0.56±0.01 (42)	0.87±0.01
+400% null edges	0.05	0	2 170 560	1.00±0	1.00±4E−4	0.95±3E−4	1.00±5E−5 (0)	1.00±3E−4 (12)	0.90±0.01
−20% edges removed	0.05	0	347 000	1.00±1E−16	0.99±7E−4	0.84±3E−4	0.99±4E−4 (0)	0.79±1E−3 (14)	0.71±0.01
−50% edges removed	0.05	0	217 056	0.98±4E−4	0.94±3E−3	0.38±2E−3	0.96±1E−3 (1)	0.65±3E−3 (34)	0.61±0.02

Supplementary Table 8: **Baseline models evaluated on central 217-column Flyvis data under three model noise regimes ($\sigma = 0$, $\sigma = 0.05$, $\sigma = 0.5$).** Five-fold cross-validation (mean ± SD). Prediction: metrics computed on noise-free data with held-out stimuli (8,000 frames). Green: value > 0.9. The Stimulus-driven MLP model cannot perform an auto-regressive rollout since it does not depend on the past neural activity at all.

condition	noise σ	params (M)	rollout steps	one-step r	rollout r	rollout RMSE
Stimulus, t_w=1, noise-free	0	7.58	—	0.42±0.29	—	—
Stimulus, t_w=4, noise-free	0	7.67	—	0.73±0.30	—	—
Stimulus, t_w=8, noise-free	0	7.81	—	0.92±0.09	—	—
Stimulus, t_w=16, noise-free	0	8.07	—	0.97±0.03	—	—
MLP, noise-free	0	29.05	20	0.95±0.04	0.96±0.05	0.080±0.057
MLP, σ=0.05	0.05	29.05	20	0.92±0.07	0.93±0.11	0.128±0.105
MLP, σ=0.5	0.5	29.05	20	0.69±0.18	0.78±0.24	0.259±0.239
EED, noise-free	0	19.52	5	0.92±0.08	0.97±0.03	0.053±0.035
EED, σ=0.05	0.05	19.52	5	0.90±0.10	0.97±0.04	0.060±0.039
EED, σ=0.5	0.5	19.52	5	0.60±0.20	0.77±0.25	0.276±0.257



Supplementary Figure 16: **Stimulus context length controls dynamical fidelity of the baseline.** For each context length $t_w \in \{1, \cdots, 16\}$, a Stimulus-driven MLP was trained to map t_w consecutive stimulus frames $\{s(t), s(t-1), \cdots s(t-(t_w-1))\}$ to the voltage of all N=13,741 neurons $v(t)$ on 64k frames of the DAVIS dataset with no noise added. We plot the Pearson-r for voltage prediction as a function of the context length t_w on the hold out test dataset over 8,000 frames. The (subsampled) points are colored by coarse cell type and the lines represent the mean performance over neurons of that type. We observe monotonically increasing performance as a function of context, and heterogeneity of performance across cell types.

H.2 Stimulus-only prediction

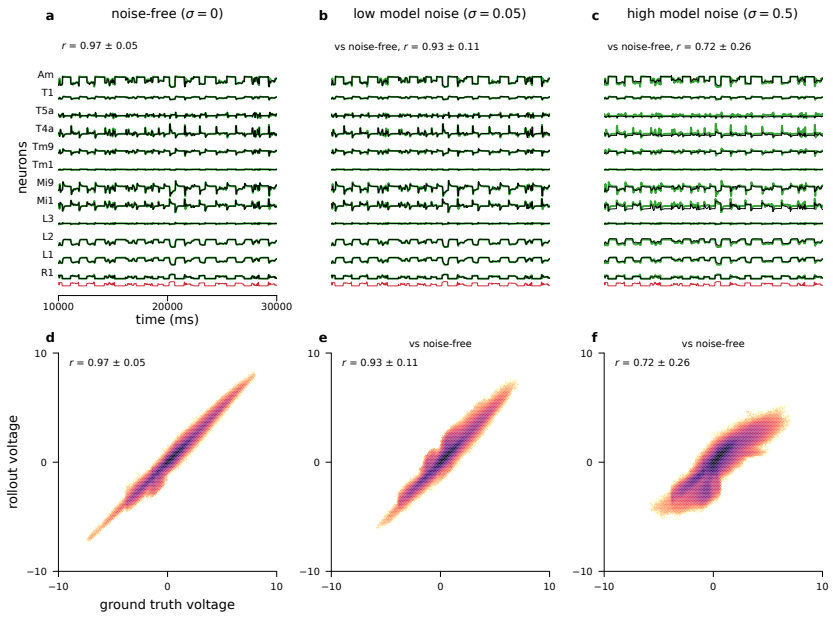
The Flyvis model optimizes a biophysically motivated, connectome-constrained model on an optical flow task, which requires the neural activities to encode past visual stimuli. The stimulus baseline model (Supp. Fig. 13) predicts the activities $v_i(t)$ of all the neurons using the input stimulus at times $t, t-1, t-(t_w-1)$, where t_w denotes a context window. The stimulus encoder MLP (2 layers, 128 hidden units) encodes each stimulus frame into a 64D latent vector and the predictor MLP (3 layers, 512 hidden units) maps the concatenated $64 \times t_w$ latents to a prediction of 13,741 neural activities. Both modules are jointly trained to minimize the MSE loss between prediction and ground truth voltage. Supp. Fig. 16 and Supp. Tab. 8 show the variation of prediction performance, quantified using Pearson's r, as a function of t_w; at $t_w = 1$ the performance is poor ($r \sim 0.4$) and increases monotonically as a function of context length reaching $r \sim 0.97$ at $t_w = 16$. This baseline demonstrates that an accurate prediction of neural activity cannot be made with the instantaneous stimulus alone.

H.3 Recurrent MLP

We built a black-box recurrent neural network using an MLP architecture (2 layers, 640 hidden units, leaky ReLU activation; $\sim$ 29M parameters) to predict the time derivative of neural activity based on the input stimulus and past activity. We begin with the noise free system, where $\sigma = 0$ in Eq. (1), and use the model

$$\frac{dv}{dt} = \mathrm{MLP}(s(t), v(t-1))\,. \tag{21}$$

We train this model by generating an auto-regressive rollout of 20 time steps and minimize the MSE loss between the predicted and ground truth voltage traces. We observed that by varying the hyperparameters of the MLP we were able to accurately capture the 1-step evolution of neural activity. However, we failed to achieve a stable roll out on validation data beyond a few hundred time steps. When measuring the MSE on an 8,000 step rollout on validation data we observed large (RMSE > 1.0) low frequency oscillations that did not go to zero even after significant increases to the training time (data not shown here). To address this, we added ResNet-style skip connections [He et al., 2015], which enable the network to easily model the low frequency oscillations in the MSE that were missed by the plain MLP. This intervention helps us learn a generalizable model for neural activity evolution and we are able to generate autoregressive rollouts of over 8,000 steps on the held-out test data with Pearson's $r > 0.90$ as shown in Supp. Tab. 8.



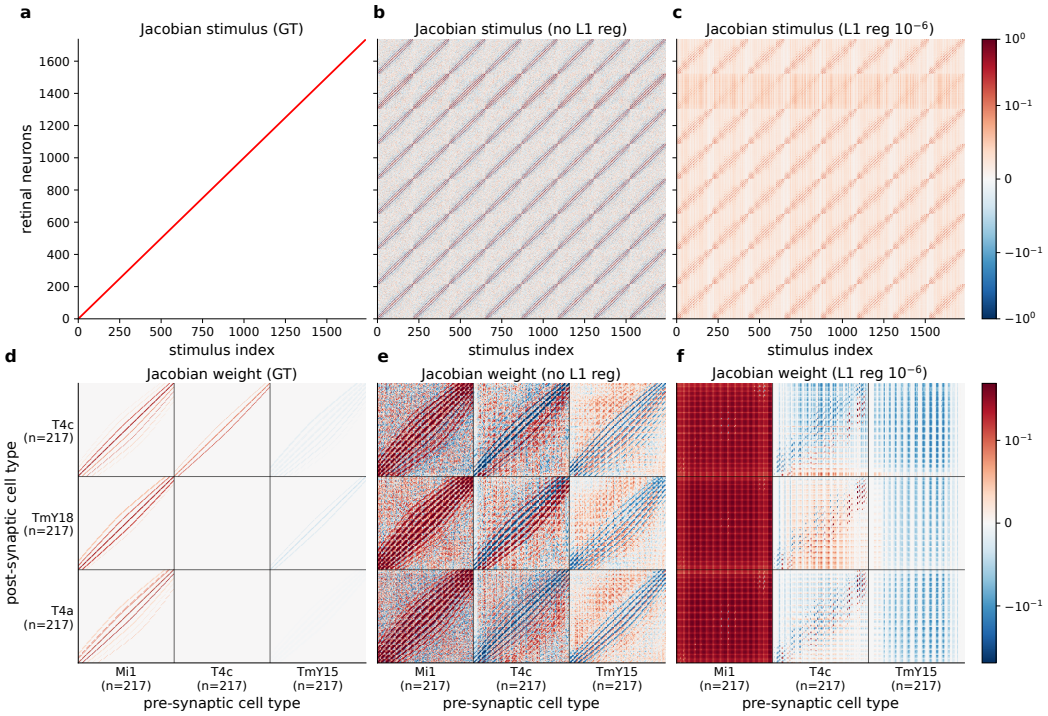
Supplementary Figure 17: **Recurrent MLP rollout prediction**: Recurrent MLP evaluated on central 217-column Flyvis data under three model noise regimes ($\sigma = 0$, $\sigma = 0.05$, $\sigma = 0.5$), tested on held-out stimuli, and compared against noise-free Flyvis simulations. Top row (**a**, **b**, **c**): rollout traces for 12 representative cell types over a 20 s window (1,000 frames at $\Delta t = 20$ ms). Green: ground-truth voltage; black: Recurrent MLP rollout prediction; red: one visual input stimulus. Bottom row (**d**, **e**, **f**): scatterplots of rollout voltage vs noise-free ground-truth voltage, pooled over all $(\text{neuron}, \text{frame})$ pairs and subsampled to $2 \cdot 10^6$ points for display. Pearson r computed on the full 8,000-frame rollout. Note the degradation in performance with the addition of model noise across columns.

Supp. Fig. 17 visualizes the noise free rollout on models trained with varying levels of noise. Unlike the GNN, the rollout performance of the Recurrent MLP is negatively impacted by increasing model noise. This is likely because with the high capacity of the model the MSE loss overfits to noise patterns in the data that do not generalize. The 20-step rollout training helps average out the noise contributions and without it the model memorizes the noise and does not generalize. It may be possible to bridge this gap with more input data that would allow for further averaging out of the noise term, changing the loss function to the likelihood under the noise model, or other architectural and training innovations.

Although the model faithfully reproduces the rollout, it does not capture the true underlying dynamics. This is best illustrated in Supp. Fig. 18 where we plot the Jacobian of the MLP update with respect to the stimuli and the pre-synaptic neuron activities, which shows a large deviation from the ground truth simulated ODE. Without the inductive bias provided by the connectome constraint, the MLP picks a different combination of stimulus and past activity to predict the next step. This agrees with the learnings from the Stimulus-driven MLP baseline, as neural activity is a specific encoding of past stimuli and therefore many such combinations could in principle exist. Adding L1 regularization does promote sparsity, and we observe that many entries in the Jacobian are pushed towards zero, but it does not help recover the *correct* Jacobian structure in the ground truth as shown in the figure.

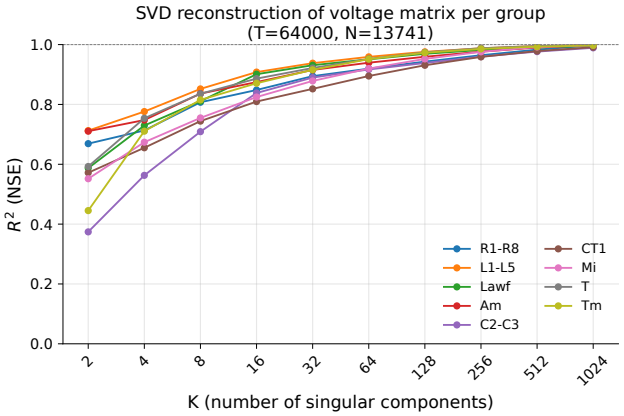
H.4 EED model

We were motivated to consider a bottle-neck architecture by the observation that the time variation of simulated neural activity matrix can be accurately reconstructed using K singular values for $K \ll N$ as shown in Supp. Fig. 19 [Churchland et al., 2012, Pandarinath et al., 2018]. We built a bottleneck architecture that we call the Encoder-Evolver-Decoder model (19M parameters), which is a neural state space model where the internal dynamics are described by an evolving latent representation, which is then read out by the measured neurons. The model architecture, depicted in Supp. Fig. 13C,

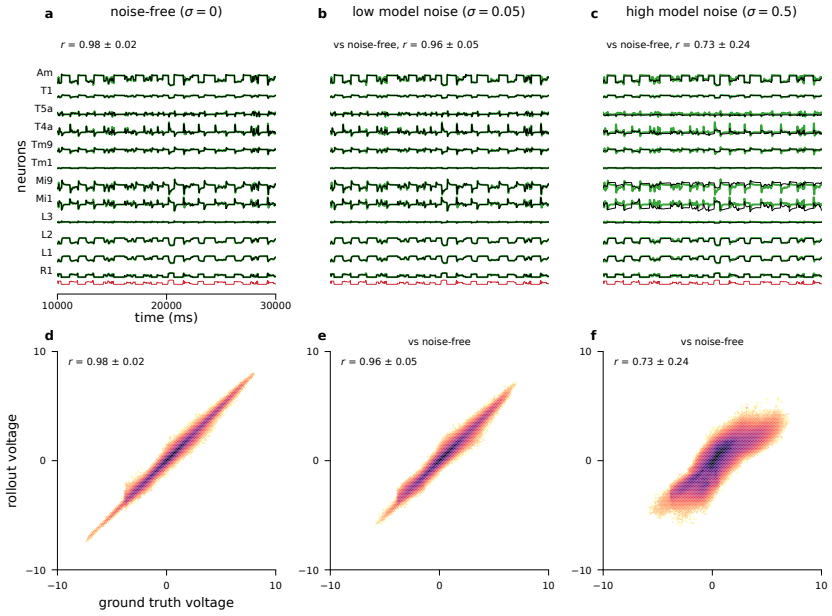


Supplementary Figure 18: **Jacobian of Recurrent MLP does not capture the true dynamics** We compute the Jacobian of the right hand side of Supp. Eq. 21 with respect to the input stimulus (top row) and the past neural activity (bottom row). For the sake of display we restrict the view to specific entries of the Jacobian that help visually distinguish the learned dynamics from the true simulation update in Eq. (1). In the top row **a-c** we show the Jacobian of the 1736 retinal neurons with respect to the 1736 input stimuli. **a** shows the ground truth, where stimulus i is fed into retinal neuron i with unit strength. **b** shows the Jacobian of the learned MLP update, which displays an interesting structure with a periodic motif repeating every 217 indices mirroring the columnar arrangement. This shows that the MLP learned a different assignment between the 1736 stimuli and the 13,741 neurons that nevertheless captures the one-step and rollout properties of the true dynamics accurately. **c** shows the impact of L1 regularization that is added to force sparsity of learned dynamics. While the true solution is also sparse, the broad L1 penalty finds a completely different sparse solution to the problem. The bottom row d-f shows the Jacobian with respect to the past neural activity, where we have restricted the view to three pre and post-synaptic cell types. **a** shows the ground truth weights from the Flyvis model, **b** shows the result from the MLP model without regularization and **c** shows the result with the addition of regularization. Again we see the same pattern as with the stimulus. Color bars in all figures use a symmetric-log normalization (linthresh $= 0.1$) so small magnitudes remain visible; each row shares one colorbar.

consists of an auto-encoder that projects to/reconstructs the observed neural activity from a 256D latent space. The visual stimulus is also encoded into a smaller 128D representation by a stimulus encoder. The time evolution operator maps the 384D concatenation of the current stimulus latent and the past neural activity latent to the next step neural activity latent. During training we perform an auto-regressive rollout of the latent state in 256D over 5 steps and use an MSE loss over the decoded activities vs the ground truth. We found that shallow MLPs (1 layer) with linear skips were sufficient to capture the dynamics in latent space. Similar to the MLP the model performance degrades with the addition of model noise. However, the EED model is able to produce stable $8{,}000$ step rollouts over test data as shown in Supp. Fig. 20. The latent bottleneck imposes a structural prior on the learned dynamics, however, this is *distinct* from the structure of the ODE imposed by the connectome graph. So while the rollout is accurately captured the Jacobian computation for the EED (not shown here) similarly misses the true structure of the simulation.



Supplementary Figure 19: **Low-rank reconstruction of the neural activity matrix.** Per-group Nash–Sutcliffe efficiency $R^2(K) = 1 - \overline{\mathrm{MSE}}_g(K)/\overline{\mathrm{Var}}_g$ for the rank-K truncated SVD reconstruction A_K of the $(T \times N)$ voltage matrix on noise-free data (T=64,000, N=13,741). For each neuron n we compute the temporal variance $\mathrm{Var}_t(A_{:,n})$ and the reconstruction error $\mathrm{MSE}_n(K) = \frac{1}{T}\sum_t (A_{t,n} - (A_K)_{t,n})^2$; both are averaged over neurons within a coarse cell-type group g (R1–R8, L1–L5, Lawf, Am, C2–C3, CT1, Mi, T, Tm). All groups exceed $R^2 = 0.99$ by K=1024 $\ll N$, motivating the Encoder–Evolver–Decoder bottleneck.



Supplementary Figure 20: **EED model rollout prediction**: EED model evaluated on central 217-column Flyvis data under three model noise regimes ($\sigma = 0$, $\sigma = 0.05$, $\sigma = 0.5$), tested on held-out stimuli, and compared against noise-free Flyvis simulations. Top row (**a**, **b**, **c**): rollout traces for 12 representative cell types over a 20 s window (1,000 frames at $\Delta t = 20$ ms). Green: ground-truth voltage; black: Recurrent MLP rollout prediction; red: one visual input stimulus. Bottom row (**d**, **e**, **f**): scatterplots of rollout voltage vs noise-free ground-truth voltage, pooled over all (neuron, frame) pairs and subsampled to $2 \cdot 10^6$ points for display. Pearson r computed on the full 8,000-frame rollout. Note the degradation in performance with the addition of model noise across columns.

I Evaluation metrics

We define here the four metrics reported throughout the paper: the one-step and auto-regressive rollout Pearson r on neural activity, and the parameter-recovery R^2 on the synaptic weights $\hat{\mathbf{W}}$, the membrane time constants $\hat{\boldsymbol{\tau}}$, and the resting potentials $\hat{\mathbf{V}}^{\text{rest}}$.

One-step Pearson r. On a held-out trajectory of T frames, at every time step $t = 0, \dots, T-2$ the model is fed the ground-truth voltage $\tilde{v}(t)$ together with the stimulus $I(t)$ and predicts the next state $\hat{v}(t+1)$. For each neuron i we compute the Pearson correlation between the predicted and the ground-truth trace,

$$r_i^{\text{1-step}} = \operatorname{corr}\big(\hat{v}_i(\,\cdot\,),\, \tilde{v}_i(\,\cdot\,)\big) \tag{22}$$

Neurons whose predicted or ground-truth trace is flat ($\sigma < 10^{-8}$) are dropped. The per-neuron correlations are then pooled in Fisher-z space,

$$\bar{z} \;=\; \frac{1}{N}\sum_{i=1}^{N} \operatorname{arctanh} r_i, \qquad \bar{r} \;=\; \tanh \bar{z}, \tag{23}$$

and we report $\bar{r} \pm (r_+ - r_-)/2$ with $r_\pm = \tanh(\bar{z} \pm \sigma_z)$ and σ_z the standard deviation of $\operatorname{arctanh} r_i$ across the N neurons.

Auto-regressive rollout Pearson r. For the rollout the model receives only the initial voltage $\tilde{v}(0)$ and the full stimulus stream $\{I(0), \dots, I(T-1)\}$; from $t = 1$ onwards it is fed back its own prediction $\hat{v}(t-1)$ to produce $\hat{v}(t)$, with no resetting to ground truth. The trajectory length T is the length of the fold's held-out stretch, 7,207 to 7,999 frames (144 to 160 s at $\Delta t = 20$ ms). The rollout trace is compared against the noise-free ground truth $\tilde{v}_i(\cdot)$ on a per-neuron basis, and the Fisher-z pooling of Supp. Eq. 23 is applied unchanged.

Synaptic-weight R^2. The recovery of the connectome is quantified by the Nash–Sutcliffe efficiency on the edge weights,

$$R^2_{\hat{\mathbf{W}}} \;=\; 1 \;-\; \frac{\frac{1}{|\mathcal{E}|}\sum_{(i,j)\in\mathcal{E}} \big(W_{ij} - \hat{W}_{ij}\big)^2}{\operatorname{Var}_{(i,j)}\big(W_{ij}\big)}, \tag{24}$$

pooled over all edges $\mathcal{E}$ of the prior graph. The numerator is the raw mean squared error, so under a linear miscalibration $\hat{W}_{ij} = a\, W_{ij} + b$ both a wrong slope $a \neq 1$ *and* a non-zero offset b inflate the residual: the metric penalises noise, scale, and bias errors. The range is $(-\infty, 1]$, with $R^2 = 1$ requiring $\hat{W}_{ij} = W_{ij}$ exactly. We additionally report the diagnostic slope a from a linear fit of $\hat{W}$ against W, which disambiguates scale miscalibration from random scatter when R^2 is low (*e.g.*Supp. Fig. 15). For models trained with extra null edges (e.g. the $+400\%$ condition) the sum runs over the augmented graph, and recovering $\hat{W}_{ij} \approx 0$ on the null edges is part of the reported number. No outlier filtering is applied to $R^2_{\hat{\mathbf{W}}}$.

Parameter R^2 for $\hat{\boldsymbol{\tau}}$ and $\hat{\mathbf{V}}^{\text{rest}}$ with outlier filtering. For the per-neuron parameters τ_i and V_i^{rest} we use the same identity-line statistic of Supp. Eq. 24, restricted to an inlier set. A neuron i is flagged as an outlier when its absolute residual exceeds a hard, parameter-specific threshold,

$$|\hat{\theta}_i - \theta_i| \;>\; \delta_\theta, \qquad \delta_\tau = 0.1, \;\; \delta_{V^{\text{rest}}} = 0.2, \tag{25}$$

in the same units as the corresponding ground-truth parameter. The thresholds are fixed once and held constant across every condition in the paper, so that outlier counts are directly comparable across rows of any cross-noise table. The filtering is motivated by how τ and V^{rest} are extracted: both are read off from the slope and offset of the learned function f_θ around the neuron's working point, and a small number of inactive or saturated neurons can produce extreme estimates that would otherwise dominate the variance in Supp. Eq. 24. The reported $R^2_{\hat{\boldsymbol{\tau}}}$ and $R^2_{\hat{\mathbf{V}}^{\text{rest}}}$ are therefore inlier-set values, with the outlier fraction $n_{\text{out}}/n_{\text{total}}$ (in percent) in parentheses next to each value, e.g. 0.94 ± 0.01 (3.3) in Supp. Tab. 4.

Cell-type clusterability (GMM accuracy, ARI, NMI). To assess whether learned per-neuron quantities recover the 65-class cell-type partition of Flyvis, we cluster each learned model feature space with a Gaussian Mixture Model and report three quantities (Supp. Fig. 7). Features are z-scored column-wise so that the high-variance $\mathbf{W}$-statistics do not drown out τ and V_{rest}, then a GMM with $K = 100$ mixture components is fit. Each neuron is assigned to its argmax component, and the predicted label set is mapped to the ground-truth cell types by the Hungarian algorithm on the $65 \times K$ confusion matrix; the GMM *accuracy* is the fraction of neurons whose Hungarian-mapped label matches the true cell type. The Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI) are computed directly between true labels and raw GMM labels and are therefore invariant to the cluster-to-type permutation. We also report k, the number of mixture components that received at least one neuron at inference time; $k < K$ when the GMM collapses redundant components (typical for the 2D embedding panel of Supp. Fig. 7c). The four candidate feature spaces are: ground-truth (τ, V_{rest}) concatenated with 8 per-neuron $\mathbf{W}$-statistics $\{\text{mean}, \text{std}, \text{min}, \text{max}\}$ of incoming and outgoing edge weights computed on the true $\mathbf{W}$ (10D); the same 10D vector built from the learned $\hat{\mathbf{W}}$; the learned 2D node embedding $\mathbf{a}_i$ used directly as input to the GNN; and the 12D concatenation of $\mathbf{a}_i$ with $(\hat{\tau}, \hat{V}_{\text{rest}}, \hat{\mathbf{W}}\text{-stats})$. UMAP is used solely to visualise the > 2D feature spaces; the GMM, the Hungarian mapping, and all reported numbers are computed in the raw feature space, never in UMAP coordinates.

Cross-validation aggregation. For five-fold CV results, every metric above is computed independently in each fold; the table value is the mean of the five per-fold values and the $\pm$ figure their standard deviation (population SD, $\text{ddof} = 0$). Pearson r values are aggregated the same way: each fold's r is the Fisher-z pooled per-neuron value of Supp. Eq. 23, and the five are averaged directly.

J Compute resources

Supplementary Table 9: **Compute resources per experiment.** GPU-hours per condition, summed across 5 cross-validation folds. All runs use a single GPU; most runs used the NVIDIA A100 SXM4 (80 GB). A "—" indicates the model was not evaluated for that condition. The *FlyVis (nominal)* row reflects the training time for most experiments conducted in this paper. We have added rows for experiments where the training time was significantly different from this nominal number.

experiment	GNN	Known ODE	Recurrent MLP	EED model	Stimulus-driven MLP (ctx=16)
Flyvis (nominal) (Tab. 1)	4.5	0.2	3	2	1
FlyVis perturbations (Supp. Tab. 4, Supp. Tab. 7)					
$+400\%$ null edges	20	1	—	—	—
$1/5$ frames	5	—	—	—	—
10% hidden	4.5	—	—	—	—
unknown stim. + SIREN recovery	33	—	—	—	—
FlyWire hybrid connectome (Tab. 2)					
flyvis e8, het. RF	4	0.25	3	2	—
flyvis e8, het. RF + n.e.	25	0.5	—	—	—
flywire l.v.f., het. RF	12.5	1	9	6	—
flywire l.v.f., het. RF + n.e.	106	5	—	—	—
agentic HPO (Supp. Tab. 6)	99	7	96	96	96